Nonstandard Educational Careers and the Returns to Vocational Education*

Moritz Mendel†

July 13, 2026

This paper examines how flexibility in educational pathways, specifically the option to enter higher education after completing a vocational program, affects students' outcomes. I estimate a dynamic model of educational choices and outcomes using administrative data from the Netherlands. The model features a choice between general and vocational education at age 16 and allows for multiple subsequent educational transitions, including entry into university after completing a vocational program. The results show that the option to enter higher education after vocational training substantially increases graduation rates and average wages. The main mechanism behind this result is that students face considerable uncertainty about their future outcomes when making educational decisions at age 16. Allowing flexibility in educational pathways enables students to adjust their choices as they learn more about their abilities and preferences, leading to improved overall outcomes.

Keywords: vocational education, sequential choices, uncertainty
JEL code: J24, J31, I26, D80

* I thank Teresa Backhaus, Manudeep Bhuller, Daniel Carvajal, Rik Dillingh, Thomas Dohmen, Hans-Martin von Gaudecker, Cormac O'Dea, Matti Sarvimäki, Paul Schäfer, Alessandro Toppeta, Paul Verstraten, Christian Zimpelmann, Maria Zumbuehl and various seminar audiences for helpful comments. I thank the CPB for hosting me for a month and for various helpful comments. I gratefully acknowledge funding by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) through CRC TR 224 (Project C01) and the Research Training Group "The Macroeconomics of Inequality". I also acknowledge funding by the Research Council of Finland (358924, 358946).

† Aalto University. moritz.mendel@aalto.fi

1. Introduction

The option to enroll in vocational training, rather than continuing with general education, offers an alternative pathway for individuals who are less academically inclined and unlikely to pursue university education. Several studies show that vocational training can improve labor market outcomes for parts of the population (McNally et al., 2022). However, expanding vocational education systems involves a fundamental trade-off. While vocational education may benefit some individuals, it may also divert others who would have had high returns from following the general track and enrolling in university. Matthewes and Ventura (2022), for example, provide evidence for this diversion effect in the UK.

This paper examines how flexibility in educational pathways, specifically the option to enter higher education after completing a vocational program, affects students' outcomes and the size of the diversion effect of vocational education. Most prior research treats the choice between vocational and general education as a static, binary decision and typically ignores subsequent educational choices. In contrast, I show that in the Netherlands, a substantial share of students who initially choose vocational education later enroll in tertiary programs. If students can enroll in university after completing vocational programs, then the extent to which vocational tracks divert students from university depends not only on how effectively the education system allocates students between general education and vocational programs, but also on how well vocational programs prepare students for higher education and on institutional barriers, such as entry requirements, that make transitions from vocational programs to university more difficult.

The opportunity to pursue university studies after completing vocational education, which I refer to as an alternative path to university, may mitigate diversion effects by allowing individuals to reverse or adjust earlier decisions. This flexibility is particularly important for students who may initially lack the grades, financial resources, or motivation to commit to university but later benefit from the availability of alternative pathways. At the same time, these alternative paths to university could exacerbate diversion effects if they encourage early choices that ultimately hinder university enrollment.

To study how the availability of alternative paths to higher education shapes individuals' outcomes, it is necessary to understand how students select into general education and vocational programs initially, how these choices shape their future options, and how students decide about university enrollment after completing a vocational program. Policies that modify either initial tracking or the availability of alternative pathways to university inevitably shift the composition of students entering higher education through each route. As a result, it becomes challenging to isolate the effect of choosing one path over another. To address this challenge, I introduce a dynamic discrete choice model in the spirit of Keane and Wolpin (1997) that features both general and vocational education and allows for nonstandard educational careers. Estimating this model enables me to analyze how initial tracking and the availability of alternative pathways to higher education jointly shape students' outcomes.

The model follows graduates of vocational school in the Netherlands[1]. Dutch students are assigned to different schooling tracks at age 12 based on test scores and teacher recommendations. Vocational schools enroll students with the lowest assessed academic potential and prepare them for vocational programs after graduation at age 16. The model begins when students graduate from vocational school and must choose between enrolling in applied high school (HAVO) or pursuing a vocational program (MBO). Individuals can enter applied university[2] either after completing applied high school, which is the direct path, or after completing a vocational program, which is the alternative path. Vocational programs take longer than applied high school and provide no explicit preparation for higher education. As a result, individuals who take the alternative path enter applied university at an older age and are potentially less academically prepared. Graduates of vocational school are close to the margin between pursuing higher education and entering the labor market after completing a vocational program, and they disproportionately come from lower-income families. Because they choose between rejoining the direct path to applied university through applied high school and pursuing a vocational program with the option of entering applied university later, they provide an ideal setting for examining the importance of alternative paths to university.

I leverage data on schooling careers, enrollment, and wage outcomes to estimate the model's parameters. While this data provides rich information on the choices and outcomes of vocational school graduates, it does not contain exogenous variation that shifts all the relevant choices and outcomes simultaneously. Differences in outcomes across students who choose different paths can be driven either by the choices themselves or by unobserved characteristics that shape both the choice and the outcome. To address this challenge, I exploit two sources of variation to identify the model. First, I use a reform to exam rules that substantially reduced the transition from vocational school to applied high school. In 2010, the Dutch government changed the exam rules in applied high school, leading to a sharp decline in the num-

[1] In particular, I focus on graduates of the theoretical branch of vocational school (VMBO-T). See 2.1 for more details.

[2] The Netherlands has two types of higher education institutions: academic and applied universities.

ber of vocational school graduates enrolling in applied high school. I exploit this reform in a difference in differences design, comparing vocational schools with high transition rates to applied high school before the reform to schools with low transition rates before the reform, to estimate its effect on applied high school enrollment and graduation, as well as on applied university enrollment and graduation. Because the reform directly shifts enrollment into applied high school, one of the pathways included in the model, it provides direct evidence on the effect of this pathway on later outcomes. The reform led to a substantial decline in applied high school enrollment among more affected schools. Schools with the highest transition rates before the reform saw enrollment fall by 9 percentage points, whereas enrollment remains constant in schools with the lowest transition rates before the reform. Yet this decline was not accompanied by any visible decline in applied university enrollment or graduation.

Second, I use cross-sectional variation in conditional choices across schools with different transition rates. While differences across schools are not themselves causal, any difference in outcomes across schools must stem either from differences in transition rules across schools or from differences in the unobserved characteristics of the students who sort into them. This observation imposes a large number of additional restrictions on the model, and thus constitutes a further powerful source of variation.

The estimated parameters show that lifetime earnings returns to applied university vary substantially across the population. For some individuals, the returns are negative because the higher earnings later in life do not compensate for the income lost while attending applied university. For most of the population, however, completing an applied university degree leads to a significantly higher lifetime income. At the same time, students face considerable dropout risk when entering applied university, in particular, individuals with low vocational school GPA have only about a 50% chance of graduating.

I run two types of counterfactual simulations using the estimated model. First, I simulate two policies that expand access to the direct path to applied university by making it easier for vocational school graduates to enter applied high school at the beginning of the model. While these policies lead to a substantial increase in applied high school enrollment, they have only a modest effect on applied university graduation rates due to the high dropout risk among compliers. Overall, these policies would have a small positive effect on wages at age 40, as they increase the number of applied university graduates and raise average labor market experience among them, since more students take the shorter direct path to applied university. The results show that easing tracking rules at the beginning of the model by expanding access to applied high school helps some students but also draws in others who are not adequately prepared, while some who would succeed in applied university at a later stage still choose vocational programs.

In the second counterfactual exercise, I simulate two policies that restrict access to the alternative path to applied university. The first policy increases the duration of applied university for people who enter after completing a vocational program by two years, and the second policy completely eliminates the option to enter applied university after completing a vocational program. Both policies are associated with substantial declines in applied university graduation rates and average wages at age 40. Removing the alternative path to university entirely would lead to a 15 percentage point decline in applied university graduation rates and a 6 % reduction in average wages at age 40.

The main explanation for the large effect of alternative paths to higher education is that many people face substantial uncertainty when deciding between vocational programs and high school at 16. Allowing them to reconsider their initial choice later in life improves outcomes significantly. One of the key findings driving this result is that the penalty for entering applied university after completing vocational training instead of applied high school is relatively small.

Finally, I compare the returns to enrolling in a higher vocational program for people with different alternative options in the baseline scenario and the scenario without an alternative path to university. This exercise is motivated by recent studies, analyzing returns to vocational education and community colleges by the next-best alternative (Matthewes and Ventura, 2022; Mountjoy, 2022), and documenting substantial negative returns to people being diverted from a more academic option. I document that the diversion effect becomes substantially larger in the counterfactual without an alternative path to university, highlighting the importance of future choices and the institutional environment in shaping long-run outcomes of graduates of vocational education.

The results of the structural model yield two crucial insights. Allowing individuals to pursue vocational training at age 16 instead of continuing high school improves outcomes for individuals who face considerable dropout risk and have only modest returns to applied university. At the same time, it diverts some individuals who would have high returns from higher education but do not yet know they want to study at 16. Keeping flexibility across educational programs at later stages allows one to reap the benefits of providing different options earlier while keeping the losses due to wrong choices under uncertainty at a young age limited.

This paper contributes to several branches of the literature. The first is a growing literature studying how vocational programs shape individuals' outcomes. McNally et al. (2022) review the literature on the returns to vocational education and training, which includes studies using a wide range of methods in different institutional settings. Silliman and Virtanen (2022) find positive wage returns to entering the vocational track for middle school graduates in Finland who choose between general and vocational education. Birkelund and van de Werfhorst (2022) find that entering the vocational track increases employment probabilities in Denmark. In contrast, Hanushek et al. (2017) show that vocational education in Germany is associated with a higher unemployment risk later in life relative to general education. Matthewes and Ventura (2022) study returns to vocational education in the UK by next-best alternative and find substantial negative returns for individuals who would otherwise enroll in university. Bertrand et al. (2021) analyze a reform of vocational education in Norway that made it easier for vocational graduates to access tertiary education. They find that the reform increased earnings among vocational graduates but had only limited effects on college enrollment. Adda and Dustmann (2023) estimate a model of wage formation for a population of German workers with vocational training and without any post-secondary education. They find that workers with vocational training accumulate cognitive-abstract skills faster, which has important consequences for their future job tasks and wages. Eckardt (2023) investigates the consequences of uncertainty in vocational program choice and quantifies the costs of occupational mismatch.

The mixed evidence in this literature highlights the importance of heterogeneity and institutional details in shaping returns to vocational education. In this paper, I estimate a fully structural model, which requires stronger assumptions but sheds light on the mechanisms that drive choices and outcomes among students entering vocational education. In addition, I explicitly account for students' future educational choices after they enroll in vocational programs. This enables me to examine how tracking rules and the availability of alternative pathways to university jointly shape later outcomes. Taken together, this approach allows me to document how the returns to vocational training vary across the population, how the expected returns to vocational versus university education depend on academic risk and ex-post wage outcomes, and how these returns are influenced by the availability of alternative routes to university.

Another closely related literature examines returns to community colleges and GED certificates in the US. Mountjoy (2022) analyzes returns to community colleges against different next-best alternatives and finds positive returns for people who would otherwise not attend college but negative returns for people who would otherwise attend a four-year college. One particularly close study is Zhu (2025), which estimates a dynamic model of education choices where individuals decide between community colleges and regular colleges and evaluates how free community college would promote social mobility. Heckman et al. (2012) analyze how the availability of the GED certificate affects high school dropout decisions and labor market outcomes. They show that the availability of the GED certificate increases dropout rates and has detrimental effects on labor market outcomes and educational attainment. This result shows that in some cases, the availability of alternative paths to higher education can have negative diversion effects, outweighing the positive effects of flexibility. While the setting in the United States is different from the one studied in this paper, many of the general dynamics are similar across the two settings.

I also contribute to a literature using dynamic discrete choice models to study the effect of education policies when people make sequential education choices under uncertainty. De Groote (2025) and Fu and Mehta (2018) use dynamic discrete choice models to study the implications of educational tracking on students' outcomes. Their work focuses on more permeable tracking systems than the binary choice between vocational and general education studied in this paper, and does not account for alternative paths to university. Stinebrickner and Stinebrickner (2012), Proctor (2022), and Arcidiacono et al. (2016) emphasize the role of learning about one's own ability via noisy signals such as grades. They find that uncertainty about one's ability drives common phenomena such as dropout or re-enrollment. While the model in this paper does not feature learning explicitly, uncertainty plays an important role in shaping people's education choices in general and the importance of alternative paths to university in particular.

A number of other studies have used additional data or variation from policy changes or RCTs to identify the incidence of unobserved heterogeneity in dynamic models of education choice. Todd and Wolpin (2023) provide a survey of structural papers that have used RCTs or quasi-experimental research designs to identify unobserved heterogeneity or to validate structural models. Joensen and Mattana (2021) use a reform to student loan policy in Sweden to estimate a dynamic model of education choice, loan uptake, and work choices. Tincani et al. (2023) use results from a randomized control trial to identify unobserved heterogeneity in a dynamic model of effort choice in high school.

The rest of the paper is organized as follows. Section 2 provides stylized facts and institutional details about the Dutch education system. Section 3 introduces a dynamic model of education choices. Section 4 introduces an identification argument and summarizes the estimation procedure. Section 5 discusses the main model results. Finally, I conclude in Section 6.

2. Setting and stylized facts

In this section, I explain relevant features of the Dutch education system, show stylized facts motivating the subsequent analysis, and summarize all the options that graduates of vocational school have.

2.1. Tracking in the Netherlands

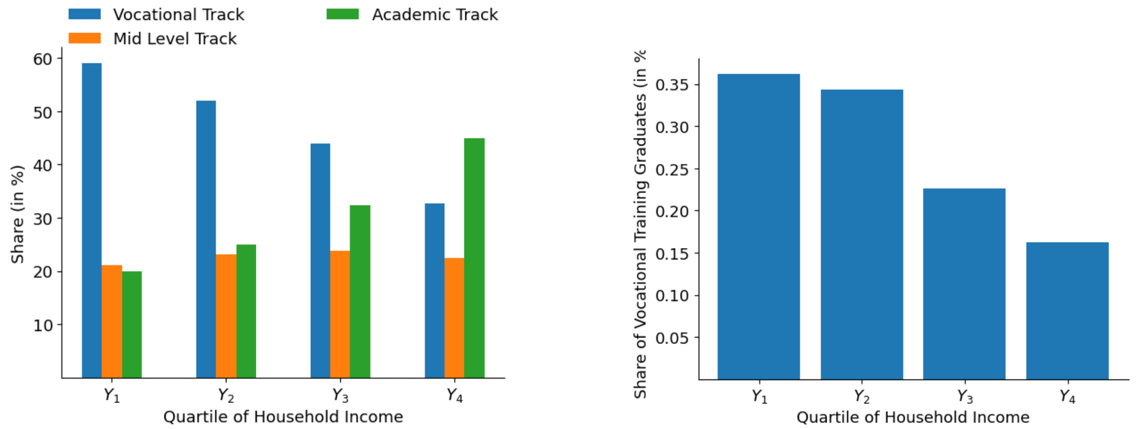
The Dutch education system separates individuals at age twelve based on grades and teacher evaluations and sends them to different secondary schools. Each school sets a different focus and prepares for a different post-secondary education. The vocational schooling track (VMBO) receives individuals with the lowest assessed academic potential, takes three years, and prepares students for vocational programs. This paper will refer to the vocational schooling track as vocational school. Vocational programs (MBO) prepare individuals for particular occupations and take two to five years. The mid-level track (HAVO) prepares individuals for applied university and takes five years. I refer to this track as applied high school in this application. Higher education in the Netherlands differentiates between applied universities, which are more practical, and academic universities. The academic track (VWO) prepares individuals for academic university and takes six years. A bachelor's degree at an applied university takes four years and a bachelor's degree at an academic university takes three years. I will describe different career options for graduates of vocational school in section 2.5. I will abstract from academic university and master's programs in this context, as most of the graduates of vocational school never enroll in either.

2.2. Data

I use Dutch administrative records to follow graduates of vocational schools. I combine information on educational careers, grades, the economic situation of their parents, school characteristics, place of residence, and future labor market outcomes. I will focus on graduates of vocational school and their future outcomes for the structural model.

2.3. Sample selection

I make several sample restrictions to make the analysis tractable.



(a) Tracking across parental income quartiles

(b) Alternative path to university across parental income quartiles

Figure 1: Descriptive statistics on tracking and alternative paths to university

Note: This figure shows descriptive statistics about tracking and the prevalence of the alternative path to university. The first panel shows the proportion of individuals entering different secondary school tracks after primary school at age 12 by parental income quartile. The figure is based on the dataset described in 2.2 and contains data from 2008-2010. The second panel shows the proportion of university graduates who have completed a vocational program before entering higher education by parental income quartile. University graduates include everyone with at least an applied university bachelor's degree. Individuals with an academic university bachelor's degree or any master's degree are also included. The figure is based on the dataset described in 2.2 and contains data from 2008-2010.

1. I remove all people who enroll in the same education program after they have already graduated from another course in the program
2. I remove all people with inconsistent education careers.
3. I remove everyone who interrupts their schooling career for more than 2 years. This is to deal with people who enroll in university late.

The first two steps remove roughly 8% of the sample, and the last step removes another 12% of the sample. The final sample for the model estimation is roughly 80% of the original sample. The difference in differences regression is based on a sample where the last step is not applied, thus relying on 92% of the original sample.

2.4. Descriptive statistics

Initial tracking: Figure 1 summarizes track choice after primary school by parental income. The figure shows that the majority of people visit a vocational school after primary school. Students from low-income backgrounds are most likely to be selected for vocational school. Track assignment is decided by teacher evaluations and a centralized test individuals take at the end of primary school. Grade differences at the end of primary school can explain a substantial part of the differences in track choice. Zumbuehl et al. (2022) show that individuals from low-income backgrounds, however, receive lower track recommendations even after controlling for grades and cognitive skills. The misallocation is thus potentially worse among individuals from low-income backgrounds than among their higher-income peers.

Alternative paths to applied university: I now consider all individuals who at least hold an applied university bachelor's degree. The second panel of Figure 1 shows the proportion of university graduates who have completed a vocational program before. The figure shows that almost a quarter of all university graduates have completed a vocational program before entering higher education. Graduates of vocational education are older and have received less academic education when they consider entering university. This shows that the alternative path to university is a well established educational pathway in the Netherlands. Conditional on reaching a tertiary degree, individuals from low socioeconomic backgrounds are twice as likely to have entered higher education after vocational training. Entering university after finishing vocational training is thus a well-established career in the Netherlands that is particularly important for individuals from low-income backgrounds.

Observed wage differences: Average wages differ substantially across educational attainment. Figure A.6 in the appendix shows the average yearly income for people with different relevant degrees. Applied university graduates show substantially larger wage growth between ages 30 and 40 than people with other degrees. People who hold an applied high school degree start out lower than graduates of vocational programs, but catch up at age 40. It is important to point out that the vocational category includes individuals with different levels of vocational training. Graduates of a higher vocational program (MBO4) are likely to earn more than other people who fall under that category.

2.5. Pathways to university

In this section, I present all possible future pathways for graduates of vocational school. From this point onward, I focus on graduates of the theoretical branch of vocational school. Vocational education is divided into four branches, and the theoretical branch admits students with the highest assessed academic ability among them. Students in this branch are closest to the margin between entering applied university and entering the labor market after completing a vocational program. Students in the lower branches are unlikely to enroll in applied university, whereas students who are initially tracked into higher academic tracks are less likely to ever attend a vocational program. The theoretical branch of vocational school is therefore the most relevant population for understanding the importance of alternative paths to university. In

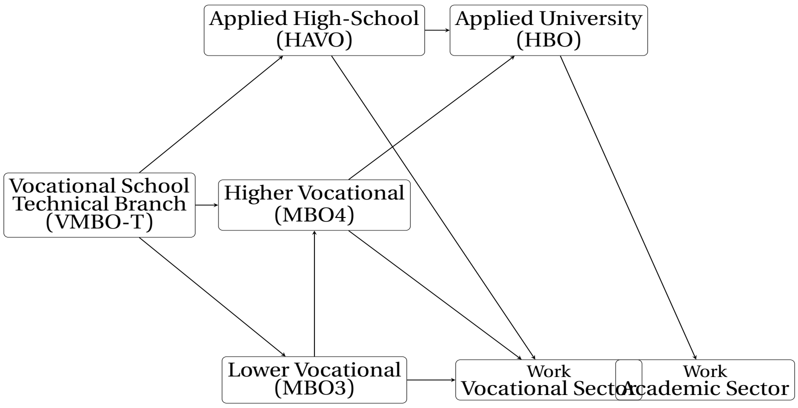


Figure 2: Pathways for graduates of vocational school

Note: This figure summarizes educational careers individuals can pursue after graduating from a vocational school.

2010, roughly one quarter of all secondary school graduates came from the theoretical branch of vocational education[3]. Consistent with the stylized facts mentioned earlier, individuals from low-income backgrounds are overrepresented in this group.

Figure 2 illustrates the possible pathways that vocational graduates can pursue after graduation. After finishing vocational school, individuals may enroll in different vocational programs or switch to the schooling track that prepares them for applied university, which I refer to as applied high school. Once individuals graduate from applied high school or a higher vocational program, they can enter applied university. If they hold a lower vocational degree, they can pursue a higher vocational degree to enter applied university in the third period. Naturally, Figure 2 includes some simplifications. In particular, I leave out possibilities that are negligible empirically. While lower vocational programs contain options beyond MBO3, most graduates of the theoretical branch choose the latter. Individuals could switch to an academic high school (VWO) after finishing applied high school (HAVO), and they could change to an academic university during their studies at an applied university. I abstract from both of these options as they are chosen infrequently. Finally, individuals can enroll in a master's degree after finishing applied university. I also abstract from this choice and treat individuals with applied university master's and bachelor's degrees equally.

Transition to applied high school: The transition to applied high school is not organized centrally. Applied high schools have employed their own rules for admitting students from vocational school (Van Esch and J., 2010). The number of individuals that transfer to applied high school from a particular vocational school thus varies by the specific rules that applied high schools in the area use and by the amount of assistance that the school offers students for their transition to applied high school.

[3] See CBS opendata

3. A model of further education

I now introduce a structural model of education. I will first introduce all the components of the model, then show how to solve the model.

3.1. Sample and decision tree

The model is based on the summary of pathways introduced in Figure 2. Individuals can first choose between a lower vocational program, a higher vocational program or applied high school. After that, they can enter applied university after applied high school or after graduating from a higher vocational program. A higher vocational program takes longer than applied high school and contains less preparation for applied university. At each stage of the model students can also decide to leave education and start working. The sample of individuals the model is estimated with consists of all graduates of the theoretical branch of vocational school, as described in the last section. In addition to the selection criteria described in Section 2, I restrict the main sample to people who graduate from vocational school between 2008 and 2010[4]. Moreover, I abstract from part-time work and only use full-time work spells to estimate wage processes.

3.2. Model organization and decision period

Individuals do not make a new decision each year. After individuals enroll in a particular education program, they stick with this decision for a potentially stochastic number of years until they either graduate or fail to do so. I choose this alternative way of specifying the model to reduce the computational complexity. A spell denotes the years an individual spends in a particular education due to their prior decision. Once the current spell is over, they make a new decision based on their current state. I thus distinguish between periods and decision periods in the model. A period $t \in \{0, 1, 2, 3..., 13\}$ denotes the number of years that have passed since the onset of the model. A decision period $\tau \in \{0, 1, ..\}$ represents the number of choices that the individual has already taken. Using decision periods allows me to substantially reduce the number of states because I do not have to include experiences for each choice in the state space.

While this model specification substantially reduces computational complexity it also introduces some limitations. In models with more granular decision periods one can

[4] The estimation process also includes estimates of a difference in differences specification that includes later cohorts. See Section 4.

endogenize dropout decisions as new information such as grades or other shocks arrives. In the Dutch administrative data I do not have information on grades for vocational programs and higher education. Moreover, it is unclear to what extent grades in vocational programs have the same predictive power as grades in high school and higher education. Lastly, the model specification that I use requires me to include several different parallel education programs. This introduces substantial computational complexity that requires me to make some simplifying assumptions.

3.3. States and fixed heterogeneity

Each individual is characterized by fixed characteristics and dynamic states. Fixed characteristics include quartile of vocational school GPA G, latent type θ, quartile of parental income Y, and school type Z. G refers to the quartile of the final GPA when people graduate from vocational school (VMBO-T) right before the beginning of the model.

School type Z refers to the school-specific transition rate to applied high school. I obtain school type Z for school j as follows:

$$Z_j = q(\frac{1}{n_j}\sum_{i \in I_j} G_i - \hat{P}_j)$$

Where $\hat{P}_j$ is the observed transition rate to applied high school in school j, n_j denotes the number of students at school j, $\frac{1}{n_j}\sum_{i \in I_j} G_i$ is the overall predicted transition rate based on vocational school GPA and parental income for all individuals i in school j, and q is a function that maps the school-specific deviation to its quartile in the distribution. It is important to point out that I do not assume that all the differences in transition rates across schools can be attributed to differences in transition rules across schools. The model both features different transition costs to applied high school across schools and allows school types Z to be correlated with unobserved characteristics θ. The school type thus captures the variation in transition rates across schools that cannot be attributed to differences in observables across schools. Latent type θ is a latent factor that captures dependence between choices and outcomes not accounted for by observed characteristics. All fixed characteristics are assigned at the beginning of the model. The joint distribution of Y, Z, and G is assigned exogenously as observed in the data. The distribution of θ depends on all the other fixed states and is estimated with all other parameters. Dynamic states include age A, current level of schooling E, and lagged choice $d_{\tau-1}$. One state is a tuple that consists of all fixed characteristics and dynamic states as described in Equation

1. Individuals start the model at age 16.

$$s_\tau = (A_\tau, E_\tau, d_{\tau-1}, G, \theta, Y, Z) \tag{1}$$

3.4. Choices and timing

Let d_τ denote an individual's choice at decision period τ. At each decision period, an individual makes a choice. Afterward, the individual stays with that choice for a potentially stochastic number of periods. After the spell is over, the individual takes the next decision.

$C(s_\tau)$ maps a state into a set of admissible choices. This function is consistent with the decision tree above. An individual who has, for example, just finished a higher vocational program can either enroll in university or leave education and work. Moreover, individuals are not allowed to enroll in the same program repeatedly. This is why the lagged choice is part of the state space. Individuals decide between applied high school, higher vocational training, lower vocational training and work in the first stage. After that, the set of choices depends on their state. People are not allowed to enroll in education programs beyond a certain age such that everyone starts working by age 32 at the latest.

If individuals enroll in a particular schooling program, they are not guaranteed to finish it. Schooling programs are associated with varying levels of dropout risk and uncertain length. Depending on their choice and the realization of academic risk, they will transition to a new stage. The stochastic function $T(s_\tau, d_{i,\tau})$ maps a state and a choice into a state at the end of the current spell.

Taking a decision thus has the following consequences. First, the transition function realizes and determines the state that an individual will end up in. Function $N(s_\tau, s_{\tau+1})$ determines all the states in between the state of departure and the state of arrival and $n(s_\tau, s_{\tau+1})$ is the number of states between s_τ and $s_{\tau+1}$. After that, the individual receives utility for each state and makes a new decision in the arrival state. Suppose the transition function, for example, determines that an individual enrolled in a higher vocational program will graduate within four years. In that case, the individual will receive utility for these four years and make a new decision after she graduates from the vocational program.

If an individual leaves education and starts working, the choice is terminal. Individuals receive the discounted lifetime income associated with their characteristics and final education.

3.5. Transitions and uncertainty

Individuals face two types of uncertainty in education: they can potentially drop out and not graduate from a particular education program, or they can graduate but with a delay.

Equation 2 shows the specification of dropout risk. $P(E_{\tau+1} = d_\tau)$ is the probability that an individual successfully graduates from the education program she enrolled in. The equations' coefficients are model objects estimated jointly with all other parameters.

$$\text{logit}\left(P(E_{\tau+1} = d_\tau \mid G, \theta, Y)\right) = \beta_{0,d}^{R} + \xi_{1,d}^{R} G + \xi_{2,d}^{R} \theta + \xi_{3,d}^{R} Y \tag{2}$$

Let min_d be the minimum number of years required to finish a degree. If an individual i completes a degree successfully, she faces a shifted Poisson process that determines the duration of her degree:

$$T_d^{E_{\tau+1}=d_\tau}(G, \theta, Y) \sim Poisson(min_d, \beta_{0,d}^{D} + \xi_{1,d}^{D} G + \xi_{2,d}^{D} \theta + \xi_{3,d}^{D} Y). \tag{3}$$

If the individual drops out, she will still spend a stochastic number of periods in the education program. The length is determined by:

$$T_d^{E_{\tau+1} \neq d_\tau} \sim Poisson(1, \beta_0). \tag{4}$$

The exact parametrization differs between the programs and can be found in the appendix. Generally I decided to parameterize the dropout and duration risk for university and applied high school more heavily than the one for vocational programs. Fully interacting each program with vocational school GPA, parental income and type would have introduced a substantial number of additional parameters. Vocational programs carry lower dropout risk in the first place which is why I decided to use a more parsimonious specification for these programs.

Agents additionally face taste shocks $\nu_{i,\tau}(d)$ to their utility. Taste shocks are modeled as an extreme value type one distribution. They are independent and identically distributed across all individuals and all choices.

3.6. Wages and nonpecuniary preferences

Wages are modeled as two separate equations for individuals with higher education diplomas and individuals without. Once students enter the labor market, they receive income for the rest of their lives. I assume that everyone works full-time after they leave school. Let k_t be work experience at time t and let E^C be an individual's combination of degrees. Log wages for the vocational sector are specified in Equation 5. Log wages in the vocational sector depend on experience, age, parental income, vocational school GPA, latent type, highest degree completed, highest degree completed interacted with experience, and experience interacted with latent type.

$$\begin{aligned} w_v(E, A_t, k_t, G, \theta, Y) = \beta^W_{0,v} + \beta^W_{1,v}E + \beta^W_{2,v}k_t + \beta^W_{3,v}k_t^2 + \beta^W_{4,v}A_t + \beta^W_{5,v}k_tE \\ + \beta^W_{6,v}k_t^2E + \xi^W_{1,v}G + \xi^W_{2,v}\theta + \xi^W_{3,v}Y + \xi^W_{4,v}k_t\theta + \epsilon_{v,t} \end{aligned} \quad (5)$$

Log wages in the academic sector are modeled separately in equation 6. I use a different specification for academic wages to allow for a flexible form of the applied university wage premium. They depend on experience, age, parental income, vocational school GPA, latent type, educational career, and an interaction between experience and latent type.

$$\begin{aligned} w_a(E^C, A_t, k_t, G, \theta, Y) = \beta^W_{0,a} + \beta^W_{1,a}E^C + \beta^W_{2,a}k_t + \beta^W_{3,a}k_t^2 + \beta^W_{4,a}A_t + \xi^W_{1,a}G + \xi^W_{2,a}\theta \\ + \xi^W_{3,a}Y + \xi^W_{4,a}k_t\theta + \epsilon_{a,t} \end{aligned} \quad (6)$$

Similar to Keane and Wolpin (1997), every choice is associated with nonpecuniary utility that is measured on the same scale as wages. I allow nonpecuniary returns $F(s, d_t)$ to depend on parental income, type, and dynamic characteristics such as experience or age. Vocational school GPA is only part of nonpecuniary rewards for applied high school, where higher grades may be associated with lower transition costs. Additionally, I include transition costs to applied high school $T(Z)$, to capture differences in transition costs across school types. I assume that the differences in rules across schools can be captured by a school-specific constant shift in the nonpecuniary utility associated with choosing applied high school in the first period. Equation 7 shows the flow utility associated with being in an education program d in state s. Since rules make only a difference for applied high school enrollment in the first period, the last term in this equation only enters for applied high school.

$$U_d(s) = F_d(s) + T_d(Z) \quad (7)$$

Equation 8 shows the utility associated with work in state s. Depending on their final education, individuals receive wages from either the vocational or academic wage equation.

$$U_w(s) = F_w(s) + e^{w(s)} \quad (8)$$

The coefficients of wage equations, nonpecuniary returns to choices, and transition costs are all model objects that are estimated. Equation 9 denotes the discounted lifetime utility from working if an agent starts working in state s at time t_s. Let $s_{w,t}(s)$ be a deterministic function that maps a state s at time t_s into the corresponding states at time t. This function updates time and experience to calculate the discounted lifetime utility from working. The term β is the discount factor and estimated along with the other parameters.

$$TU(s) = \sum_{t=t_s}^{T} \beta^{t-t_s} U_w\big(s_{w,t}(s)\big) \tag{9}$$

3.7. The agent's problem and solution algorithm

Expected utility is the weighted average over all possible paths a decision could lead to. One needs to sum over all states that could be reached from a particular state choice combination. Let $R(s_\tau, d_{i\tau})$ be the range of potential outcomes one can reach from state s_τ and decision $d_{i\tau}$ and let $P_{s_\tau,d_{i\tau}}(s_{\tau+1})$ be a probability distribution over the range of outcomes. Equation 10 shows the optimization problem of an individual in the model at state s_τ. Note that if the decision is to work then $n(s_\tau, s_{\tau+1})$ is zero and the whole term is equal to $TU(s)$ as specified in Equation 9.

$$\max_{d \in C(s_\tau)} \left\{ \sum_{s_{\tau+1} \in R(s_\tau,d)} P_{s_\tau,d}(s_{\tau+1}) \left[\sum_{s \in N(s_\tau,s_{\tau+1})} \beta^{n(s_\tau,s)} U_d(s) \ + \ \beta^{n(s_\tau,s_{\tau+1})} V(s_{\tau+1}) \right] \ + \ \nu_{i,\tau}(d) \right\} \tag{10}$$

I solve the model by backward induction. Let $V(s)$ be the expected continuation value from reaching state s, let $V(s, d)$ be the expected continuation value from choosing d in state s, and let $V(s, d, \hat{s})$ be the expected continuation value of choosing d in state s and reaching $\hat{s}$. I refer to a state as terminal if the only choice available is work. At the last time period where people can take decisions, all states are terminal. To solve this model, I proceed as follows. I start with the highest age at which agents can make decisions in the model and repeat the following steps for all ages down to age 16.

1. Collect all possible state choice combinations (s, d) of age t

2. For all combinations of states and work choices, assign the terminal utility

$$V(s, w) = TU(s)$$

3. For all non-terminal combinations:

   a) Collect all reachable states $\hat{s} \in R(s, d)$ and their probability $P_{s,d}(\hat{s})$

   b) Collect the expected continuation value from reaching $\hat{s}$: $V(\hat{s})$

c) Now combine the expected continuation value with the flow utility on the path from s to $\hat{s}$:

$$V(s,d,\hat{s}) = \sum_{\widetilde{s}\in N(s,\hat{s})} \beta^{n(s,\widetilde{s})} U_d(\widetilde{s}) + \beta^{n(s,\hat{s})} V(\hat{s})$$

d) Get the continuation value of (s,d) by taking the expected value over $\hat{s}$:

$$V(s,d) = \sum_{\hat{s}\in R(s,d)} P_{s,d}(\hat{s}) V(s,d,\hat{s})$$

4. Now get $V(s)$ by getting the expected value of the maximum of $V(s,d)$: $V(s) = E[max\{V(s,d)\}] = \sigma log(\sum_d e^{\frac{V(s,d)}{\sigma}})$ where σ is the scale of the extreme value taste shocks.

3.8. Initial states

During the model simulation I need to assign initial states to individuals. I first assign all observed states based on a probability distribution that I directly estimate from the data. Thereafter I obtain a latent type for each simulated individual. The distribution is parameterized with a multinomial logit specification equal to:

$$\log \frac{P(\theta_k \mid G_i, Y_i, Z_i)}{P(\theta_K \mid G_i, Y_i, Z_i)} = \alpha_k + \beta_k^{\theta,1} G_i + \beta_k^{\theta,2} Y_i + \beta_k^{\theta,3} Z_i \tag{11}$$

This specification allows for the school type to be arbitrarily correlated to the latent type.

4. Estimation and identification

This section will summarize the estimation procedure and the variation I use to identify the model parameters. In the first part of this section, I will first describe a difference in differences specification that I use to estimate the effect of a reform to exam rules in applied high school on applied high school enrollment and subsequent outcomes. This variation is directly informative about the selection problem in the model and is thus an important pillar of the simulated method of moments strategy used to estimate the model. Thereafter I will describe cross-sectional variation across different schools and how this variation can be informative for model parameters even if the comparison is not causal. In the last part, I will present the estimation strategy and describe how the moments jointly identify the components of the model. Finally

I present the model fit and show that the model is able to reproduce the observed moments well.

4.1. Exam rule reform

This section presents the results of a reform to exam rules in upper secondary school. In August 2010, the Dutch government introduced a series of reforms to exams in upper secondary schooling[5]. The reform was rolled out in two steps. Starting with the 2011/2012 school year exam rules required students to achieve an average grade of at least 5.5 in the central exam. Starting with the 2012/2013 school year exam rules in applied high school required students to score no lower than a 5 in at most one of three core subjects, Math, Dutch, and English, with the other two graded at 6 or higher. Following this reform, transition rates from vocational school to applied high school declined substantially.[6] The most plausible explanation is that schools became more reluctant to admit vocational school graduates, fearing lower completion rates under the stricter exam requirements.

To analyze the effect on applied high school enrollment, I estimate a difference in differences specification comparing students who graduated from vocational schools with high pre-reform transition rates to applied high school against students from schools with low pre-reform transition rates. Schools decide whether to allow students to transition from vocational school to applied high school after graduating at grade 10, and I document that they vary substantially in their transition rates. Some schools allow more than 30% of students to transition to applied high school, while others allow only around 10% (See figure A.3). Policy reports confirm that schools had very different transition policies, with some considerably more lenient than others (van Esch and Neuvel, 2009). According to Van Esch and J. (2010), one factor contributing to this variation is whether a school has an applied high school track on site. Schools with high transition rates either have a more lenient transition policy themselves or have a nearby school offering applied high school with a lenient policy.

To the extent that schools respond to the reform by adjusting their transition policies, the effect should be larger for schools with high transition rates, since they have more room to adjust. Other schools already had strict screening mechanisms in place and thus have less room, and less reason, to change their policies. I compare schools with low and high transition rates in a difference in differences specification to estimate the effect of the reform on applied high school graduation and subsequent outcomes. Schools are grouped using the same procedure as in the model estimation,

[5] See here for the original text specifying changes in rules. See here for a text specifying the rollout
[6] See Figure S.2 in the supplemental appendix.

based on the covariate adjusted transition rate to applied high school between 2008 and 2010 (see Section 3.3).

The year 2011 serves as the first reform year. The policy was approved in August 2010, too close to the start of the school year for schools to substantially adjust their transition policies for the 2010/2011 cohort. I use schools in the bottom half of the transition rate distribution as the control group and schools in the top half as the treatment group, further distinguishing between schools just above the median and those above the 75th percentile. In Section S2 of the supplemental appendix, I also report results using the 25th percentile as the cutoff for the control group, with the three upper quartiles treated as individual treatment groups.

Parallel trends assumption The key assumption is that trends between different groups are parallel in the absence of the reform. One potential concern is that the drop in transition rates to applied high school was driven by other factors, such as changes in student composition across schools. Figure A.2 shows the percentage difference of several observable characteristics between treatment groups and control groups before and after the reform. The figure shows that differences across treatment and control groups are modest and stable over time. Since people are initially tracked into different schools at age 12, which is four years before the reform, it is unlikely that many students would be able to adjust their school choice in anticipation of the reform.

Throughout the sample period, the minimum wage was adjusted on several occasions, with two notable adjustments occurring in 2023 and 2024. These reforms could introduce differential trends across school types as these minimum wage reforms have different impacts depending on the wage structure of an industry or region. Another consequential reform that took place later was the reform of the student finance system in 2015, which reduced the amount of grants that students receive in higher education. All the groups were similarly affected by the reform and parental income is well balanced across school types, so it is unlikely to drive differential trends across school types. I discuss the potential impact of both reforms on the results when presenting the difference-in-differences estimates.

Results Formally, I estimate the specification shown in Equation 12. Z_i refers to the school type analogue to the definition in Section 3.3. t_i refers to the cohort year of the individual. The coefficient β_3 captures the effect of the reform on the outcome variable Y_i. I use heteroskedasticity robust standard errors in the estimation of the effects. I am not including any further controls in the specification to keep the effect as close as possible to the model specified earlier.

$$Y_i = \beta_0 + \beta_1 \mathbb{1}\{t_i \geq 2011\} + \beta_2 Z_i + \beta_3 Z_i \mathbb{1}\{t_i \geq 2011\} + \epsilon_i \tag{12}$$

Figure 3 shows event study plots comparing the treatment and control groups over time for four outcomes: graduation from applied high school, enrollment in applied university, graduation from applied university within nine years, and log wages at age 28.

The figure confirms that the reform led to a large and significant decline in applied high school enrollment among graduates of vocational schools that initially had higher transition rates, while schools with lower transition rates show no significant change. Schools in the highest quartile of pre-reform transition rates experience a decline in applied high school enrollment of approximately 9 percentage points relative to the control group, while schools in the second highest quartile experience a decline of approximately 4 percentage points. This distribution of enrollment changes across school types supports the idea that the reform to exam rules in applied high school should have a larger effect on schools with initially higher transition rates, since these schools have more room to adjust their transition policies. Applied high school graduation declines by around 4 percentage points for schools in the highest quartile of pre-reform transition rates and around 2 percentage points for schools in the second highest quartile.

Both applied university enrollment and graduation are largely unchanged by the reform. This is a striking result, given that the reform shifts a substantial number of people out of applied high school. Applied university enrollment and graduation are noisier than the applied high school margins, and both series show more fluctuation in the pre-reform period. A small decline in applied university enrollment and graduation could in principle be masked by this fluctuation, but the figures nonetheless show no substantial break in trend for either outcome, even as applied high school enrollment declines by around 9 percentage points for schools in the highest quartile of pre-reform transition rates.

Applied university graduation is flat for both control and treatment groups between 2009 and 2013 which suggests that the 2015 reform of study financing had no substantial effect on applied university graduation in our sample as it should have created a break in the trend at some point during this time period. This is because the later people graduate from vocational school the more likely they would be to enter applied university only after the reform was implemented in 2015. There was a small dip in applied university enrollment in 2012 and 2013 for the control group. To the extent that the 2015 reform only affects the control and not the treatment group at all, which is not plausible as the majority of people should be equally affected across schools, this dip would mask a decrease in applied university enrollment in the treatment group by around 1 percentage point.

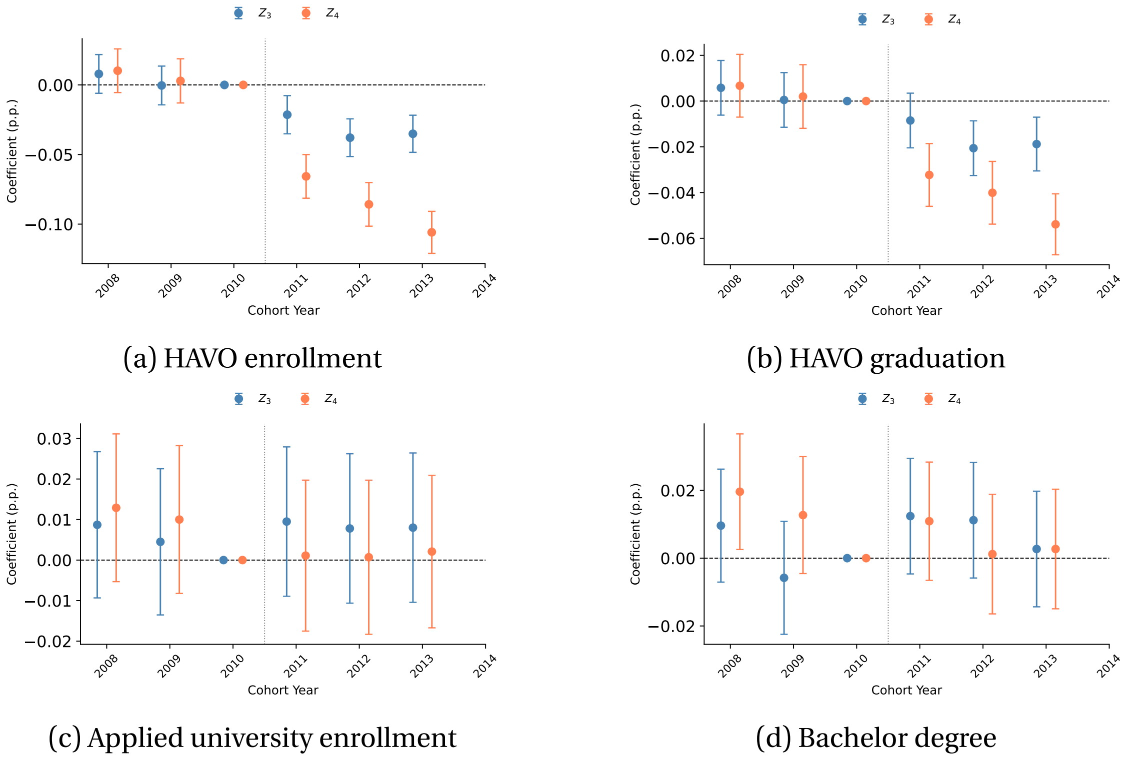
The overall pattern, then, is that the reform had a large effect on applied high school enrollment and graduation but hardly any effect on applied university enrollment and graduation. This pattern could be driven either by marginal applied high school enrollees not being particularly likely to have entered applied university in the first place, or by students who are pushed out of applied high school taking the alternative path into applied university instead.

Finally, Figure A.1 shows the effect of the reform on wages at age 29. Wages increase slightly after the reform, though the change is not significant for the school type with the highest pre-reform transition rates. It is worth noting that two large minimum wage changes occurred in 2023 and 2024, which could introduce differential trends across schooling groups. In particular, the fact that the wage increase is larger for the second highest quartile than for the highest quartile suggests that these differential trends in wages are not driven by the reform itself but by other factors. Overall, the estimated wage effects of the reform on the most affected school type is consistent with the story that the reform had a limited effect on downstream outcomes, such as overall education attainment or future wages.

Relationship to the model The parameters from the difference in differences specification connect directly to one of the key selection problems in the model. It is unclear to what extent observed differences between people who enroll in applied high school and those who enroll in vocational programs reflect selection or genuine treatment effects of the programs themselves. The difference in differences design lets me isolate a treatment effect of applied high school enrollment on subsequent outcomes for part of the population, which I use as a direct source of identification. To exploit this variation, I incorporate the reform into the model itself, allowing schools to adjust their transition policies in response to it. Specifically, I introduce a reform variable R, equal to one if an individual is subject to the post-reform policy, and add it to the transition cost function $T(Z)$, which is now parameterized as:

$$T_{ahs}(Z, R) = \beta_0^T Z + \beta_1^T ZR \tag{13}$$

These reform coefficients are estimated jointly with the remaining model parameters. For each candidate parameterization, I compute the model implied difference in differences result and compare it to the difference in differences estimates obtained from the actual reform. I restrict this comparison to schooling outcomes and use wage effects as untargeted moments.



(a) HAVO enrollment

(b) HAVO graduation

(c) Applied university enrollment

(d) Bachelor degree

Figure 3: Event-study estimates of the KV reform

Note: Each panel shows difference-in-differences coefficients for the interaction of cohort year with school type, relative to the 2010 reference cohort (see Equation 12). Z_1 and Z_2 (pooled) are the control group; Z_3 and Z_4 are the treatment groups and have progressively higher baseline transition rates to applied high school. Bars show 95% confidence intervals. The vertical dotted line marks the reform cutoff (2010/2011). Applied university graduation is measured 10 years after graduation from vocational school. See Appendix Figure A.1 for the corresponding wage effects.

4.2. Cross-sectional variation across schools

As described in Section 2.5, the transition from vocational school to applied high school differs substantially across schools. Figure A.3 shows that the lowest school type sends around 10 % of its graduates to applied high school, while the highest school type sends more than 35 %. While there is some evidence that schools use different rules when deciding whom to admit to applied high school, it is unclear to what extent differences in transition rates are driven by differences in rules $T(Z)$ or selection across schools $corr(Z, \theta)$. Conditional choices and outcomes across school types, however, provide information that helps to inform the model estimation. The argument is that the differences across school types add a timing restriction to the model. Differences in downstream outcomes across school types are either due to differences in choices in the first period which were induced by rule differences or due to differences in unobserved characteristics across school types. This restriction allows me to add many additional restrictions to the estimation procedure while only adding a small number of additional parameters to the model.

Figure A.8 shows conditional outcomes across school types. The figures show that conditional on not choosing applied high school in the first period people from schools with lower initial transition rates have higher wages and higher graduation rates from applied university. For each parameter candidate there must be a correlation between latent types and school types and school specific transition costs such that the model is consistent with these observed differences in conditional choices and outcomes across school types.

4.3. Simulated method of moments

This section contains a description of the full estimation procedure. I use indirect inference to estimate 194 parameters $\hat{\theta}$. Equation 14 shows the criterion function. I select the parametrization that minimizes the weighted squared distance between the specified set of moments computed on the observed M_D and the simulated data $M_S(\theta)$. I weight the statistics with a diagonal matrix W that contains the variances of the observed moments[7] (Altonji and Segal, 1996). I use a package for the estimation of scientific models by Gabler (2022) for the optimization of the criterion function[8].

$$\hat{\theta} = \arg\min_{\theta \in \Theta} (M_D - M_S(\theta))W^{-1}(M_D - M_S(\theta))' \tag{14}$$

Description of moments: Table 1 provides an overview of all 524 statistics used in the model estimation. The moments combine a battery of observed statistics on enrollment, graduation, and wages. Additionally they contain both cross sectional variation in conditional choices and outcomes across school types and the difference in differences estimates of the effect of the 2010 reform to exam rules in applied high school on applied high school enrollment and subsequent outcomes.

The enrollment percentage for a particular program indicates how many people have been enrolled in that program. Enrollment percentages are included for each quartile of parental income, for each quartile of vocational school GPA and each combination of school type and quartile of vocational school GPA. The final degree combination indicates all degrees an individual receives before starting work. If a person first graduates from a vocational program and then graduates from an applied university, her degree will be higher vocational & applied bachelor's. Final degree combinations are included for each quartile of parental income and each quartile of vocational school GPA. Furthermore, I include quartiles of last schooling age for each combination of parental income quartile and final degree, for each combination of

[7] I adapt some of the entries of the weighting matrix. In particular I increase the weight of school type variation and did coefficients because they are economically important sources of variation. I also increase the weight on wage quartiles as this category contains relatively few but economically important moments. Finally I reduce the weight on age moments.

[8] I use a global version of the BOBYQA algorithm within the package (Powell et al., 2009).

vocational school GPA quartile and final degree, and for each combination of school type, vocational school GPA quartile, and applied high school enrollment status. The last schooling age is when an individual is done with education and starts to work. Since I do not allow re-enrollment, there is always one age at which individuals leave education. In practice, I allow individuals to take a gap of up to two years between spells, which will be part of the degree duration. Wage quartiles over time are wage quartiles for individuals by final degree at ages 30, 35, and 40. Conditional choices across school types include the proportion of individuals enrolling and graduating from applied university across school types, conditional on their choices in the first period.

Additionally, I match the coefficients of six separate wage equations. Let T^u denote the years someone needs to finish applied university. Equation 15 is estimated on a panel that includes all full-time individuals who left school without a bachelor's degree from the third period onward. Equation 16 is estimated on a panel that includes all full-time individuals who left school with a bachelor's degree from the sixth period onward. Both equations capture how wages depend on observable states featured in the model. Both include work experience k_t, vocational school GPA G, and parental income Y. Equation 15 additionally includes the non-academic degree of an individual and an interaction between the non-academic degree and work experience. This is either a higher vocational degree (MBO4), a lower vocational degree (MBO3), an applied high school degree (HAVO), or no degree after vocational school (VMBO-T). Equation 16 additionally includes a fixed effect for all non-university degrees that individuals have completed before entering university. Furthermore, it includes the years an individual took to finish her bachelor's degree. Both of these equations suffer from selection bias. Since the model explicitly models the selection process, they are still helpful for identifying wage components.

$$W_{v,t} = \alpha_{v,0} + \alpha_{v,1}E + \alpha_{v,2}k_t + \alpha_{v,3}k_t^2 + \alpha_{v,4}k_tE + \delta_{v,0}G + \delta_{v,1}Y + \omega_{v,t} \tag{15}$$

$$W_{a,t} = \alpha_{a,0} + \alpha_{a,1}E^C + \alpha_{a,2}T^u + \alpha_{a,3}k_t + \alpha_{a,4}k_t^2 + \delta_{a,0}G + \delta_{a,1}Y + \omega_{a,t} \tag{16}$$

Finally there are four equations that capture how wages depend on school type across different choice trajectories. Equation 17 is estimated on a cross-section of all full-time individuals in period thirteen. The equations contain school type fixed effects and fixed effects for quartiles of vocational school GPA.

$$W_h = \alpha_{h,0} + \alpha_{h,1}Z + \alpha_{h,2}G + \alpha_{h,3}Y + \omega_h \tag{17}$$

Finally I match coefficients of Equation 12. I use enrollment and graduation in ap-

Table 1: Summary of moments used in the estimation.

Type of Moment	Number
I. Program enrollment by parental income, vocational school GPA	32
II. Degree combination by parental income, vocational school GPA	56
III. Last schooling age quartile by parental income and degree, vocational school GPA and degree	120
IV. Life-cycle wage quartiles by education	45
V. Coefficients of wage equations	29
VI. Program enrollment by school type and vocational school GPA	64
VII. Conditional choices across school types	46
VIII. Conditional wages across school types	28
IX. Last schooling age by school type, vocational school GPA and applied high school enrollment	96
X. Did Coefficients of the 2010 reform	8

Note: This table summarizes all 524 moments used to estimate the model. The left column indicates a particular category of statistics, and the right column indicates the number of moments the respective category has.

plied high school and applied university as outcomes. The inclusion of a reform specific transition cost in the model as specified in Equation 13 allows me to match the difference in differences estimates of the reform to exam rules in applied high school.

Identification The main challenge in identifying the model is that differences in outcomes and choices across educational programs can be due to differences in unobserved characteristics across individuals or due to effects of the programs themselves. Selection based on unobserved characteristics is modeled with latent types θ in the model. The extent to which the model can produce correct counterfactual predictions thus depends on how well the distribution and incidence of latent types is identified and estimated. I will first explain how the observed moments inform the model parameters and then explain how parameters from the difference in differences specification and cross-sectional variation across school types help to identify the distribution of latent types.

The observed information on enrollment, graduation, and wages provides important information for wage parameters, dropout risk and utility parameters. Parameters of the wage equation are informed by the auxiliary wage regressions and the wage distribution at different ages for different schooling types. Information on enrollment and graduation rates informs dropout and duration risk. Utility parameters are informed by average deviations from the choices that maximize monetary rewards.

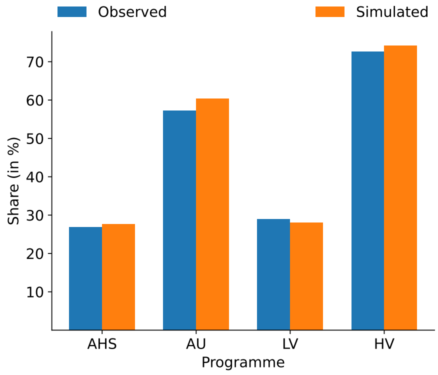
Finally, latent types and taste shocks introduce unobserved heterogeneity across people in the model. Taste shocks introduce transitory heterogeneity across people while latent types introduce permanent heterogeneity. Both of these categories are informed by the observed moments because the distribution of enrollment choices, graduation and outcomes holding observed background variables fixed provides information on residual heterogeneity. The estimation procedure will choose the distribution of latent types and taste shocks that best matches the observed moments. Consider as an example a situation where in one group of people 50% choose to enroll in applied high school and 50% choose a lower vocational program after which they dropout and work instead of continuing education. This pattern would be consistent with a model where people in the group have different latent types. On the other hand a situation where choices among people in the group are more balanced and there is a substantial number of people who enroll in applied university after a higher vocational program and some people who do not enroll in applied university after applied high school would be more consistent with a model where taste shocks play an important role.

The results of the difference in difference specification provide a direct source of identification for the distribution and incidence of latent heterogeneity. The reform has changed the composition of students in applied high school which allows me to derive treatment effects of applied high school enrollment on subsequent outcomes for a part of the population. By including these effects, the estimation procedure will choose latent types and parameters in a way such that the treatment effect on the treated of shifting a similar amount of people out of applied high school will correspond to the results of the 2010 reform.

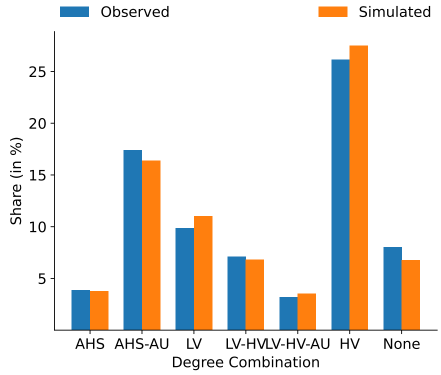
The variation across school types is a second source of variation that helps to identify the distribution of latent types. For any candidate distribution of latent heterogeneity that is consistent with observed data and the difference-in-differences estimates of the reform there must exist a distribution of latent types across schools and school specific transition costs to applied high school such that the model is consistent with the observed differences in conditional choices and outcomes across school types.

4.4. Model fit

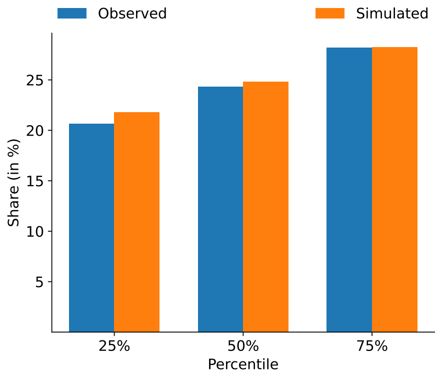
Figure 4 briefly summarizes the model fit. A more detailed summary can be found in Section S3 in the supplemental appendix. The first two panels show the fit of enrollment proportions and degree combinations for individuals with high vocational school GPA. Both sets of simulated moments are closely aligned with their observed counterparts. The third and fourth panels show wage quartiles for individuals with



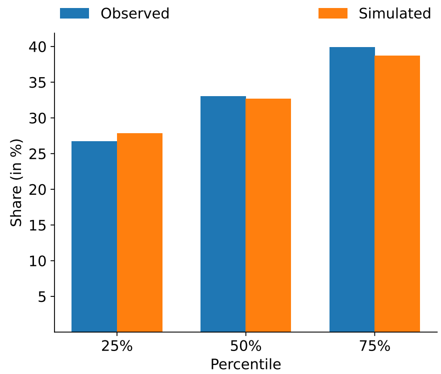
(a) Enrollment proportion



(b) Degree combination



(c) Wage quantiles for bachelor's degree holder age 30



(d) Wage quantiles for bachelor's degree holder age 40

Figure 4: Summary of model fit

Note: This figure summarizes the model fit. The figures compare observed moments based on the dataset described in 2.2 and simulated moments from a model with the estimated parameters. The blue bars show the observed moments, and the orange bars show simulated moments. The x-axis labels for the figures in the first row correspond to education programs specified in Figure 2. AU refers to applied university, AHS refers to applied high school, HV refers to higher vocational program, and LV refers to lower vocational program. Labels in the second figure represent paths through the decision tree specified in Figure 2. AHS-AU, for example, indicates that an individual graduates from applied high school first and from an applied university after that. The figures in the second row depict wage percentiles for individuals who hold a bachelor's degree at ages 30 and 40. In particular, they show the 25th, 50th, and 75th percentiles of wages among all individuals who work full-time and hold an applied university bachelor's degree.

an applied university degree at ages thirty and forty. The simulated moments closely match the observed moments. Section S3 in the supplemental appendix also shows the model fit of the difference in differences estimates of the 2010 reform to exam rules in applied high school. The simulated effects align well with the estimated coefficients. The untargeted simulated wage effect of the reform are also close to the estimated coefficients.

5. Results

I now present the empirical findings of the paper. First, I discuss the estimated distribution of wage returns to applied university and dropout risk at applied university. After that, I show several counterfactual simulations derived from the estimated model.

5.1. Mechanisms

The estimated model contains information about the distribution of wage returns to applied university and the distribution of dropout risk at applied university. Both of these factors are crucial in understanding how policy choices shape students' outcomes in the setting of this paper. A detailed list of parameter estimates and standard errors can be found in Section S1 of the supplemental appendix.

Distribution of observed and unobserved characteristics: Individuals are characterized by parental income, vocational school GPA, and a latent type. The joint distribution of parental income and vocational school GPA is observed in the data. The distribution of latent types conditional on parental income and vocational school GPA is estimated along with the other model parameters. Figure 5 shows the joint distribution of fixed characteristics in the model. Consistent with differences in earlier track assignment by socioeconomic status, there are more people from low-income households than people from higher-income households. Latent types are correlated with observed characteristics. Individuals in the lowest quartile of vocational school GPA are more likely to have type θ_5, whereas individuals in the highest quartile of vocational school GPA are more likely to have type θ_1. Differences in outcomes across individuals with the same parental income, vocational school GPA, and latent type are only due to different realizations of random shocks and are not systematic[9]. If we thus compare wages of individuals with the same parental income, vocational school GPA, and latent type but different education levels, the resulting returns correspond to the average causal wage difference across the education levels for this subgroup. Wage returns conditional on all fixed characteristics are thus average returns to university for all individuals in the respective subgroup. I can thus assign average wage returns to different degrees and average dropout risks in different programs to each individual in the sample.

Wage returns to applied university: The model reveals substantial heterogeneity in wage returns to an applied university degree relative to a higher vocational degree. While average returns at age 40 are positive across the population, they vary sharply by individual characteristics and latent type.

To quantify returns to applied university, I compute the average gap in discounted lifetime income between individuals with an applied university bachelor's degree and those with a higher vocational degree for each combination of observed

[9] The model features school type as an additional fixed characteristic, but since it only affects the utility associated with choices in the first period, it does not directly affect lifetime outcomes.

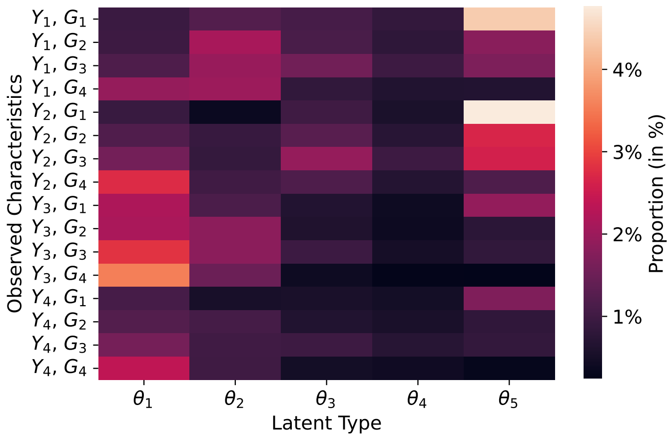


Figure 5: Distribution of fixed characteristics in the model

Note: This figure shows a heatmap visualizing the distribution of fixed characteristics in the model. The vertical axis represents one combination of vocational school GPA and parental income, while the horizontal axis represents one latent type each. Both vocational school GPA and parental income are exogenously given to the model. The distribution of the latent type given parental income and vocational school GPA is estimated by the model.

characteristics and latent type. To construct lifetime earnings, I use the estimated discount rate of around 5% and assume that people receive zero income when they are not working. Furthermore, I assume that wages are constant between the ages of 40 and 70. The results should be interpreted keeping in mind that these assumptions could lead to a downward bias of lifetime earnings differences if wages continue to grow faster for applied university graduates after age 40.

Figure 6 displays the distribution of lifetime returns. Returns differ most by latent type and much less by parental income or vocational school GPA. The first latent type experiences negative lifetime returns to applied university across most characteristics, whereas the fifth type gains substantially. Figure A.5 shows the same distribution measured at age 40, rather than over the life cycle. At this age, returns are positive for all groups but remain heterogeneous, ranging from roughly 15% for the second latent type to more than 50% for the fifth. For some individuals, the earnings premium later in life is insufficient to offset early career income losses from delayed labor market entry. One of the main mechanisms in the model that creates substantial returns for applied university degrees is that graduates of applied universities receive substantially larger returns to labor market experience than individuals holding other degrees. The model abstracts from job and program-specific factors, such as field of study, or occupation. These omissions may explain why latent type plays such a large role in shaping lifetime income differences across education levels. It is nonetheless remarkable that wage returns do not substantially vary across people with different vocational school GPAs.

Figure A.4 shows the returns to an applied university degree relative to all other

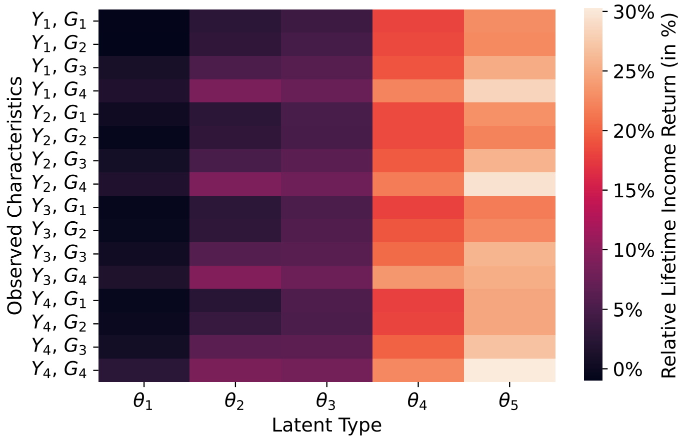


Figure 6: Distribution of lifetime earnings returns to applied university

Note: This heatmap summarizes the distributions of returns to applied universities. The vertical axis represents one combination of vocational school GPA and parental income, while the horizontal axis represents one latent type. The returns are expressed in percentages. The returns are obtained by calculating the difference in discounted lifetime income of individuals with applied bachelor's degrees and of individuals with higher vocational degrees for each combination of observed characteristics and latent type. Observed characteristics are parental income and vocational school GPA. School type is not included since it has no direct effect on wages. Discounted lifetime income differentials within a group of observed characteristics and latent type are average returns to applied university for all individuals in that group since wages don't systematically differ conditional on these variables. The value of the respective group is then assigned to each simulated individual to obtain a distribution.

degrees. These returns are larger than when the alternative is a higher vocational degree, but the distributional patterns are similar. Returns rise when all other degrees are considered as alternative options, since a higher vocational degree is associated with higher wages than a lower vocational degree or dropping out after middle school. Graduates of higher vocational programs also earn slightly more than those with only an applied high school degree, although this advantage fades over time.

Dropout risk: Heterogeneous dropout risk is the main source of variation in outcomes across individuals in the model. Given that the model predicts substantial returns to applied university for many individuals, a key question is what factors prevent graduation among those without an applied bachelor's degree. Parameter estimates indicate that differences in dropout risk at applied university[10] play a particularly important role. To describe how dropout risk is distributed, I calculate the dropout probability at applied university for each combination of observed characteristics and latent type in the model.

Figure 7 shows the resulting distribution. Individuals in the lowest quartile of vocational school GPA face substantial dropout risk and graduate with a probability of about 50%, while those in the highest quartile of vocational school GPA graduate with a probability of around 80%. These differences are consistent with the large gaps in

[10] The other education programs outlined in Figure 2 also involve dropout risk. I focus on applied university in this section because it is the most relevant program for the long-run outcomes of vocational school graduates. See the supplemental appendix for dropout risk in other programs.

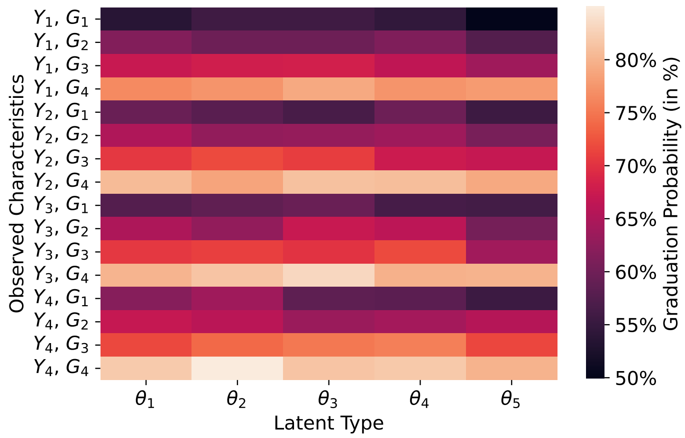


Figure 7: Distribution of graduation probability at applied university

Note: This figure shows a heatmap visualizing the probability of graduation at applied university. The vertical axis represents one combination of vocational school GPA and parental income, while the horizontal axis represents one latent type. This figure is obtained by calculating the number of applied university students who drop out for each combination of observed characteristics and latent type in the model. Observed characteristics are parental income and vocational school GPA. School type is not included since it has no direct effect on wages. Observed dropout rates within a group of observed characteristics and latent type represent academic risk for all individuals in that group since academic risk does not systematically differ conditional on observed characteristics and latent type. The value of the respective group is then assigned to each simulated individual.

observed dropout rates across quartiles of vocational school GPA. High dropout probabilities for students with low vocational school GPA highlight the importance of providing strong non-academic outside options. The estimated parameters indicate that although students entering university from vocational education are slightly more likely to drop out than those coming from applied high schools, the difference is not particularly large. This finding is noteworthy because it suggests that completing a vocational program that does not explicitly prepare students for applied university and emphasizes practical skills does not substantially reduce the likelihood of succeeding at an applied university.

Unobserved factors also shape dropout risk. Individuals with high expected returns to applied university are also more likely to graduate. Understanding which individuals are affected by a given policy is therefore crucial. If reforms shift mainly those with modest returns and high dropout risk into applied university, the resulting wage effects will be substantially smaller.

5.2. Counterfactuals

In this section, I examine how two policy margins shape students' long-term educational outcomes. The first margin concerns the initial tracking decision, where students enter either an applied high school or a vocational program. The second margin involves the flexibility to pursue higher education after vocational training, specifically the option to enroll in an applied university following a vocational program.

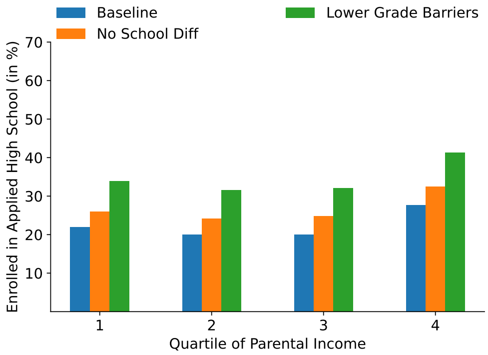
These two policies interact. If the main benefit of the alternative path to applied university is that it lowers the cost of entering applied university, then relaxing the initial tracking constraint may be a more effective reform than maintaining a separate alternative route to university. However, if students who later use the alternative path were never close to choosing applied high school initially, preserving access to university through vocational education becomes essential for improving their long-run outcomes. To evaluate these mechanisms, I conduct two sets of counterfactual simulations. First, I simulate policies that expand admission to applied high schools. Second, I simulate policies that restrict access to applied universities for students completing vocational programs. Comparing results across these scenarios isolates the relative importance of early tracking versus flexibility in later educational choices.

Transition costs: Not everyone who graduates from vocational school can easily enroll in applied high school. Applied high schools have different transition rules, so depending on a student's school and vocational school GPA, it can be harder to enroll. I change two aspects of the model to understand how a more flexible tracking system would affect outcomes. First, I remove differences across school types and simulate a world where everyone faces the rules of the most lenient schools. Second, I reduce transition costs for individuals with lower vocational school GPAs because these students face more barriers when moving to applied high school. Reducing transition costs for these students represents a world with fewer formal barriers to entering applied high school. It is important to point out that the counterfactual predictions of the model are not accurate if parents can react by changing the earlier track assignment of their children. In the Netherlands track assignment at that time, however, was mostly based on teacher recommendations and standardized test scores, which are not easily manipulated by parents.

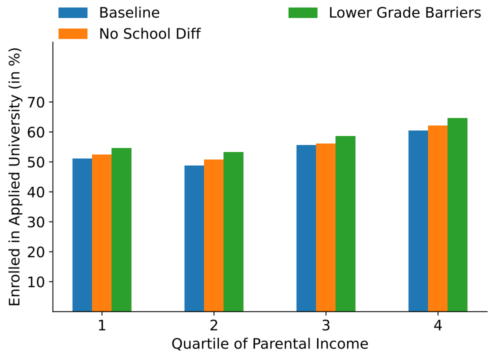
Consistent with the reduced form findings earlier both counterfactuals increase applied high school enrollment but have a limited effect on applied university graduation. Figure 8 shows how program enrollment and graduation change across quartiles of parental income. The first panel shows that removing school differences increases applied high school enrollment by about 5 percentage points, while additionally lowering barriers for people with lower vocational school GPA increases it by about 15 percentage points across all income groups. The second panel shows smaller effects on applied university enrollment. The first counterfactual raises enrollment by about 1 percentage point, and the second raises it by between 2 and 5 percentage points. The third panel shows that the rise in applied high school graduation is much smaller than the rise in enrollment, especially in the second counterfactual. The fourth panel shows that applied university graduation increases by less than 1 percentage point in the first counterfactual and by about 2 to 4 percentage points in the second.

There are two reasons why the counterfactuals do not substantially increase applied university enrollment, even though they substantially raise applied high school enrollment. First, the counterfactuals shift many students into applied high school who would have pursued the alternative path to university in the baseline scenario. Second, the counterfactuals move students with a high dropout risk at both applied high school and applied university into applied high school. Figure A.7 shows how outcomes vary across quartiles of vocational school GPA, which is particularly important for determining dropout risk in the model. The figure indicates that the first counterfactual mainly shifts students with high vocational school GPA into applied high school. Many of these students would have pursued the alternative university path in the baseline scenario, which explains why the increase in applied university graduation is small. The second counterfactual raises applied high school enrollment across all groups of vocational school GPA. However, individuals with lower vocational school GPA experience a smaller increase in applied high school graduation, indicating that a substantial share of them drops out of applied high school.

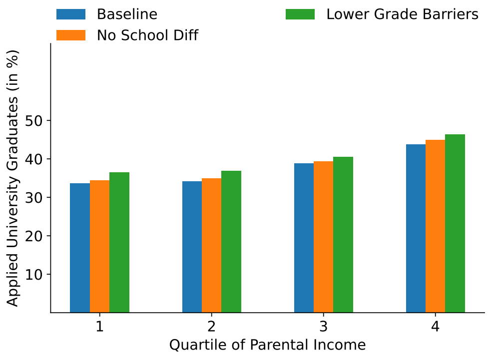
Figure 9 shows how average wages change across the two counterfactuals. Removing school types increases lifetime wages by about 0.4%, and reducing barriers for people with lower vocational school GPA increases them by about 0.6%. The wage effects at age 40 are noisier, ranging between -0.1% and 0.5% for the first counterfactual scenario, and between 0.2% and 1.2% for the second. Both policies move some individuals from a higher vocational degree to an applied university degree, and shift others from the vocational path onto the direct path toward university. Because the direct path is shorter, part of the wage increase comes from the greater labor market experience that bachelor's degree holders accumulate as a result. The fact that the wage increase from the first counterfactual is more than half that of the second counterfactual for some subgroups, even though its effect on applied university graduation is small, points to this being an important mechanism. Another mechanism behind the increased wages is that the second counterfactual shifts some individuals from work into applied high school in the first period. These results show that easing the transition from vocational school to applied high school can improve economic outcomes for some students, but would not lead to substantial wage increases across the population, even though the policies considered affect a substantial share of it. This finding is broadly consistent with the reduced form results in Section 4. The wage effects from the simulated counterfactuals take into account long run wage outcomes, while the estimated reform effect relied on wages at age 29. Taking into account the



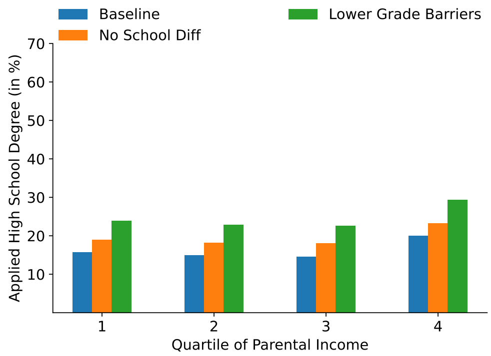
(a) Applied high school enrollment



(b) Applied university enrollment



(c) Applied university graduation



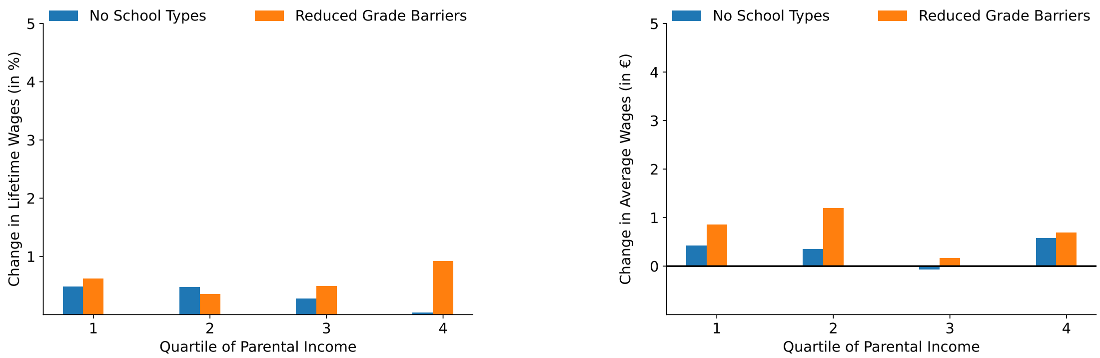
(d) Applied high school graduation

Figure 8: The effect of enforcing higher acceptance rates at high school

Note: This figure shows how enforcing higher acceptance rates at applied high school would affect enrollment and graduation at applied university and applied high school. Blue bars show the baseline model's proportions of applied university and applied high school graduates and enrollees. The orange bars show the proportions in the counterfactual scenario where all schools behave like the most lenient schools. The green bars show the proportions in the counterfactual scenario where people with lower vocational school GPA face lower barriers to enter applied high school, and where all schools behave like the most lenient schools. The proportions are shown for each quartile of parental income. Notably, most individuals graduating from vocational school are from households in the lower-income quartiles.

uncertainty with which wage effects are estimated, the effect of the counterfactual that removes school types is nonetheless consistent with the reduced form estimates.

Removing school types raises lifetime wages by increasing graduation and labor market experience and would benefit most affected students. Reducing barriers for people with lower vocational school GPA has larger effects on graduation and wages, but these gains must be weighed against higher dropout rates. Most people who drop out of applied high school will enroll in a vocational program, and some will graduate from applied university via the alternative path later in life. This pattern shows how vocational programs and the alternative path to university provide a safety net for students who struggle in applied high school, which dampens the effects of increased dropout rates in applied high school. Expanding access to applied high school helps some students but also draws in others who are not prepared, while some students who would succeed in applied university at a later stage still choose to enroll in vocational programs.



(a) Lifetime wages

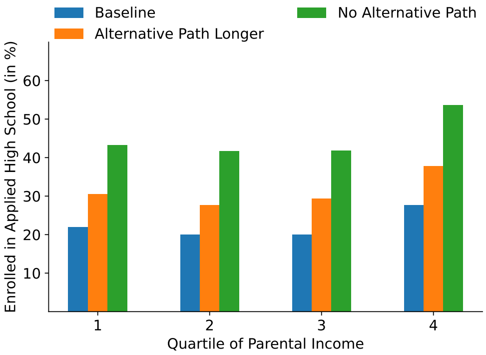
(b) Wages at age 40

Figure 9: Wage effect of enforcing higher acceptance rates at high school

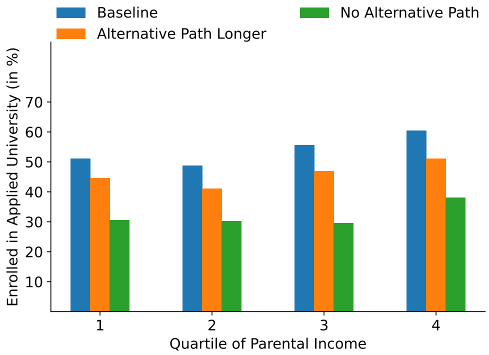
Note: This figure shows how enforcing higher acceptance rates at applied high school would affect lifetime wages and wages at age 40. The first panel shows changes in average lifetime wages, and the second panel shows changes in average wages at age 40. The blue bar shows wage changes in a counterfactual scenario where each school behaves like the most lenient schools. The orange bar shows wage changes in the counterfactual scenario where people with lower vocational school GPA face lower barriers to enter applied high school, and where all schools behave like the most lenient schools. The changes are shown for each quartile of parental income. Notably, most individuals graduating from vocational school are from households in the lower-income quartiles.

Vocational path to university: In this section, I run two counterfactual simulations to understand the importance of the alternative path to university for wages and schooling outcomes. The alternative path to university allows people who did not enter applied high school initially, either because they failed to comply with admission requirements or because they were not interested initially, to enter applied university. The first motive should disappear if the initial tracking decision were to be made more leniently, while the second motive, being a fundamental consequence of uncertainty in the model, should remain. Thus, I simulate two policies where I restrict access to the alternative path to university while making the initial tracking decision more lenient in the same way as last section. First, I simulate a policy where the alternative path to university takes two years longer than the direct path. This counterfactual is inspired by schooling systems where people need to go back to school to receive some form of high school degree before entering university after finishing a vocational program. The second counterfactual scenario removes the alternative path to university entirely. It is important to point out that the effect of increasing the length of the alternative path would constitute a lower bound if increasing the length would also substantially change the dropout rate from the program.

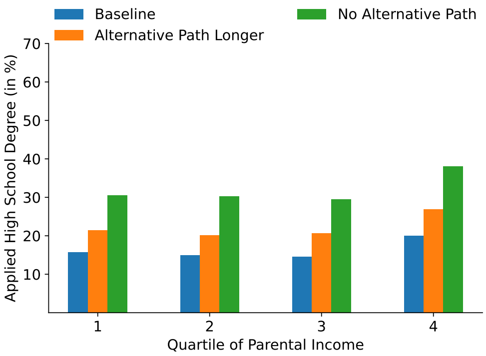
Both counterfactuals would lead to a substantial decrease in applied university enrollment and graduation rates. Figure 10 shows how the counterfactual scenarios affect enrollment and graduation at applied high school and applied university across parental income groups. In the first counterfactual, where the alternative path to university is lengthened by two years, applied high school enrollment increases by around 5 percentage points, and in the second counterfactual, where the alternative path to university is removed, applied high school enrollment increases by around 10



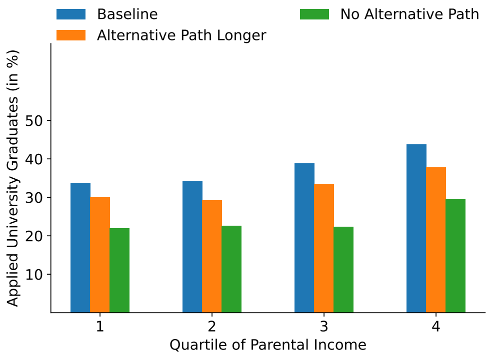
(a) Applied high school enrollment



(b) Applied university enrollment



(c) Applied high school graduation



(d) Applied university graduation

Figure 10: Effect of having no vocational path to applied university

Note: This figure shows how restricting access to the alternative path to university would affect both enrollment and graduation at applied high school and applied university. Blue bars show the baseline model's proportions of applied university and applied high school graduates and enrollees. The orange bars show the proportions in the counterfactual scenario where the alternative path to university is lengthened by two years. The green bars show the proportions in the counterfactual scenario where people are not allowed to enter applied university after graduating from a higher vocational program. The proportions are shown for each quartile of parental income. In both counterfactual scenarios entry to applied high school is made more lenient according to the counterfactual in the previous section, that removes school types and barriers for people with lower vocational school GPA.

percentage points across income groups. Despite the large increase in applied high school enrollment and graduation, both counterfactuals would lead to a substantial decrease in applied university enrollment and graduation. Increasing the length of the alternative path to university by two years would lead to a decrease in applied university graduation rates of around 5 percentage points, and removing the alternative path to university entirely would lead to a decrease in applied university graduation rates of about 10 percentage points. In both counterfactuals, the impact of restricting access to the alternative path to university thus dominates the effect of making the initial tracking decision more lenient.

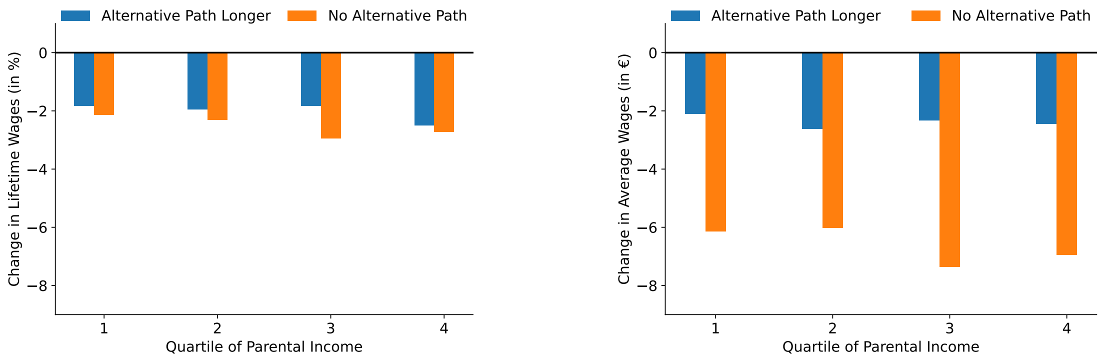
Consistent with the enrollment and graduation effects, both counterfactuals would lead to a substantial decrease in average lifetime wages. Figure 11 shows that both counterfactual simulations would reduce average lifetime income by around 2% across income groups. Removing the vocational path entirely, however, has a much larger impact on wages at age 40, which fall by more than 6%, compared to a 2% decline in the first counterfactual scenario where the program length is increased by two years. The effect on wages at age 40 is larger when removing the alternative path because this leads to a larger reduction in applied university graduation rates and shifts a substantial number of individuals with high expected returns out of applied university. This negative wage effect is partially offset by the fact that many people are shifted into applied high school who would have taken the alternative path in the baseline scenario. As a result, the overall impact on lifetime wages ends up being similar across the two counterfactuals.

The results of the two counterfactual simulations show that restricting access to the alternative path to university would have substantial negative effects on wages and applied university graduation which cannot be offset by making the initial tracking decision more lenient. Making the tracking system more lenient fails to select a substantial number of people who would benefit from entering applied high school after finishing vocational school. Another pattern that helps to explain the importance of alternative paths to university is that even people who drop out of applied high school enter applied university via the alternative path later in life. These patterns highlight that some people are unprepared to succeed in academic education earlier in their lives, while being able to successfully graduate from applied university later. In such cases, it is impossible to perfectly select people into applied high school at age 16, which provides a rationale for maintaining alternative paths to higher education as fundamental options in the education system.

Taking all of the results together yields three important insights. First, the option to pursue a vocational program improves outcomes for many people who face large dropout risk and modest wage returns to applied university. Second, while it is possible to improve outcomes by making the initial tracking decision more lenient, it is, however, difficult to perfectly differentiate between individuals who would benefit from entering applied high school and those who would not at this stage. Finally, the option to pursue the alternative path to higher education allows people to correct earlier choices after they finish their vocational program. The alternative path to university thus helps to mitigate the negative effects of tracking at age 16 while keeping the benefits of offering different education options.

5.3. Returns to vocational education

Past research has analyzed the returns to vocational education for individuals with different outside options in a fixed institutional setting. Using the model estimated in this paper, I can examine how institutional features influence the earnings effects of vocational education along different margins. In particular, I simulate an expansion



(a) Lifetime wages

(b) Wages at age 40

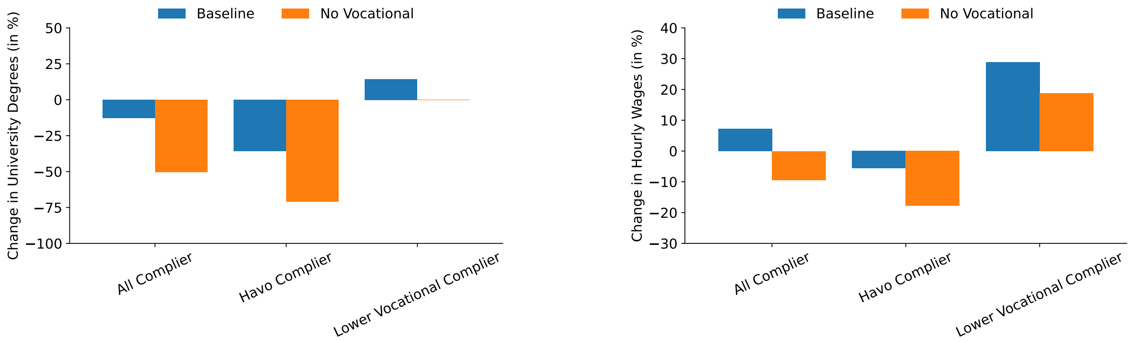
Figure 11: Wage effect of having no vocational path to applied university

Note: This figure shows how restricting access to the alternative path to university would affect lifetime wages and wages at age 40. The first panel shows changes in average lifetime wages, and the second panel shows changes in average wages at age 40. The blue bars show wage changes in the counterfactual scenario where the alternative path to university is lengthened by two years. The orange bars show wage changes in the counterfactual scenario where people are not allowed to enter applied university after graduating from a higher vocational program. In both counterfactual scenarios entry to applied high school is made more lenient according to the counterfactual in the previous section, that removes school types and barriers for people with lower vocational school GPA. The changes are shown for each quartile of parental income. Notably, most individuals graduating from vocational school are from households in the lower-income quartiles.

of the higher vocational program under both the baseline scenario and a scenario without an alternative path to university. I then compare how outcomes change for individuals who would have chosen a different option in the absence of the expansion in each scenario.

To understand the mechanisms, Figure 12 shows how the expansion affects educational outcomes and wages in each scenario. All compliers are individuals whose initial educational choice changes in response to the expansion. HAVO-compliers are individuals who would have chosen the direct academic path to university in the absence of the expansion and instead choose the higher vocational option when the expansion is introduced. This margin corresponds to the diversion margin commonly studied in the literature. MBO3-compliers are individuals who would have chosen the lower vocational program in the absence of the expansion and instead move to the higher vocational program under the expansion. In many previous settings, this margin corresponds to individuals without post-secondary education, but here the relevant outside option is another vocational program due to the more differentiated structure of vocational education in this context.

The figure shows that without the alternative path to university, both complier groups experience substantially lower applied university graduation rates and wages at age 40. In particular, the return to choosing the higher vocational program becomes substantially more negative for HAVO-compliers, indicating that the presence of an alternative path to university can mitigate diversion effects. Similarly, the wage gain for MBO3-compliers is smaller in the scenario without an alternative path to univer-



(a) Applied university graduation

(b) Wages at age 40

Figure 12: Returns to higher vocational program by alternative choices and policy scenarios

Note: This figure shows how returns to a higher vocational program by different alternative choices would change without alternative path to higher education. The figure is obtained by simulating an expansion of the higher vocational program both in the baseline scenario and the scenario without alternative path to university. The expansion is implemented in the model by increasing the utility associated with enrolling in a higher vocational program. HAVO-compliers refer to people who would have chosen applied high school in the absence of the expansion. MBO3-compliers refer to people who would have chosen the lower vocational program in the absence of the expansion. The first panel shows the change in university graduation rates among the different complier groups. The second panel shows the change in average hourly wages at age 40 among the different complier groups. The y-axis represents the change in percentage points for both panels. On the x-axis, there are two bars for each of the complier groups. The orange bar shows the return to choosing a higher vocational program in the baseline scenario, while the blue bar shows the same return in the scenario without alternative path to university.

sity.

The results show that the returns to vocational programs depend crucially on the institutional setting and the available outside options across different margins.

6. Conclusion

In this paper, I have investigated how flexibility in educational choices affects students' outcomes. I estimated a dynamic model of education choices that follows individuals after they graduate from vocational school in the Netherlands. The results show that returns to applied university differ across the population but are substantial for most individuals. I also find that many students face a substantial dropout risk at applied university, creating a wedge between enrollment and graduation rates. Counterfactual simulations show that promoting the direct path to applied university by easing the transition from vocational school to applied high school would benefit some students but would not generate large increases in overall graduation rates or wages. Restricting access to the alternative path to applied university, however, would substantially reduce both graduation rates and wages. The results of this study highlight the importance of both variety in educational options and flexibility in educational pathways for improving students' outcomes in an environment where students are heterogeneous and face uncertainty about their future outcomes.

A. Appendix

A.1. Reduced form: wage effects

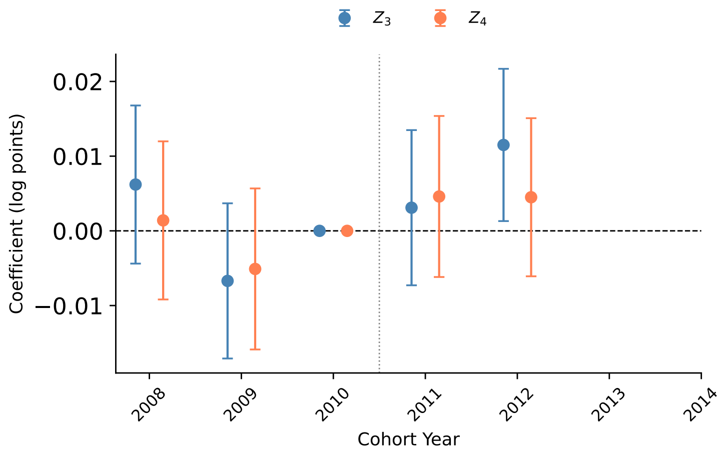
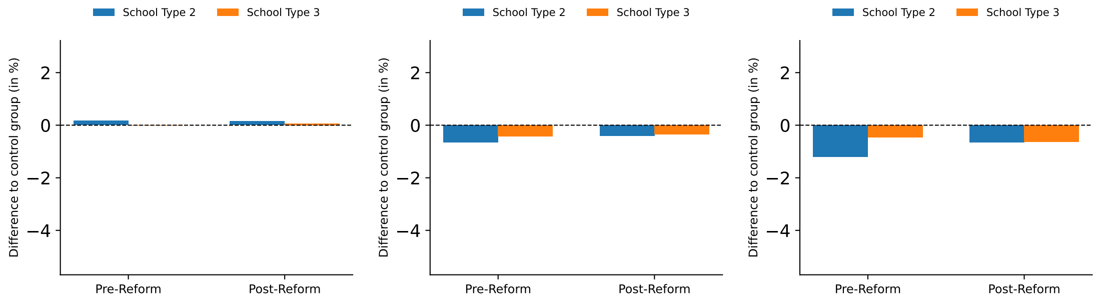


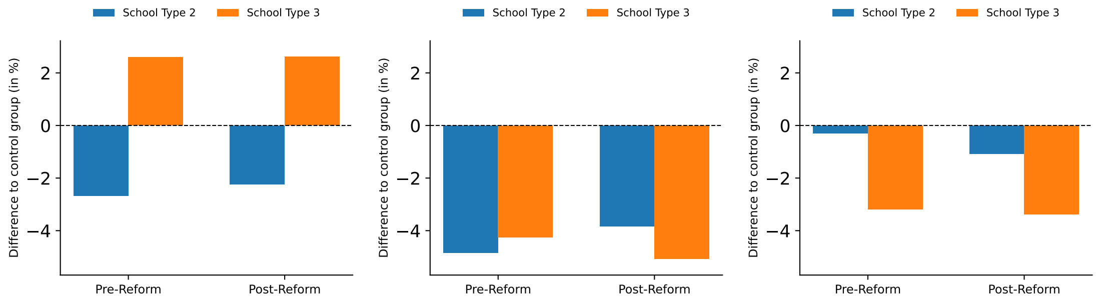
Figure A.1: Event-study estimates of the KV reform on log wages (age 29)

Note: Difference-in-differences coefficients for the interaction of cohort year with school type, relative to the 2010 reference cohort (see Equation 12). Z_1 and Z_2 (pooled) are the control group; Z_3 and Z_4 are the treatment groups and have progressively higher baseline transition rates to applied high school. Bars show 95% confidence intervals. The vertical dotted line marks the reform cutoff (2010/2011).

A.2. Robustness: covariate balance



(a) Age (b) Combined Test Score (c) Household Income



(d) Distance to School (e) Dutch Background (f) No. of Exam Subjects

Figure A.2: Covariate balance: treatment minus control differences, pre- vs. post-reform

Note: Each panel shows the relative difference in means, $(\bar{X}_T - \bar{X}_C)/\bar{X}_C$, expressed as a percentage, for a given covariate, with the pre-reform and post-reform cohorts shown side by side. Control here pools school types 0 and 1; treatment groups are school types 2 and 3. All panels share the same y-axis scale so relative magnitudes are comparable across covariates. The dashed line marks zero. Population density is excluded pending further checks.

A.3. Model parametrization

In this section I show the full model parametrization. Wage equations have been specified in 5 and 6 respectively.

Nonpecuniary returns Formula 18 shows nonpecuniary utility for working without applied university degree. Utility for working with applied university degree looks the same without the degree term.

$$F_v(Y, A_t, E) = \beta^F_{0,v} + \beta^F_{1,v} E_t + \beta^F_{2,v} A_t + \xi^F_{0,v} Y \tag{18}$$

Formula 19 shows nonpecuniary utility for applied university and both forms of vocational training. Utility returns to applied high school additionally include grades.

$$F_d(Y, \theta) = \beta^F_{0,d} + \xi^F_{0,d}\theta + \xi^F_{1,d} Y \tag{19}$$

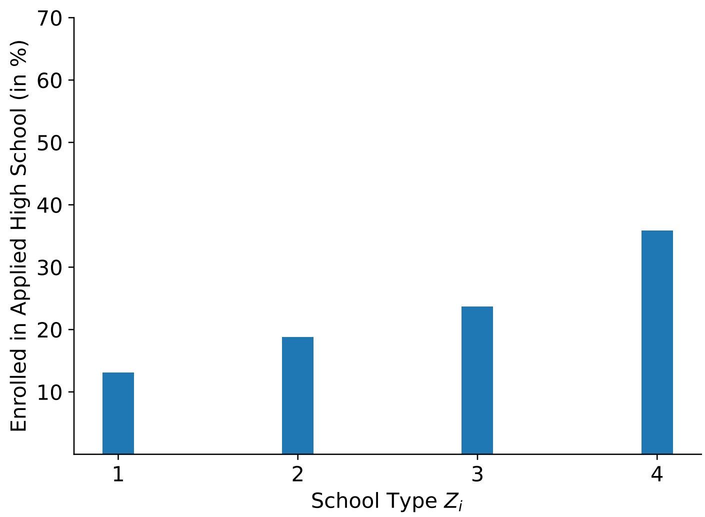
Dropout risk Formula 2 shows the specification that holds for applied high school.

For university I additionally include an indicator whether an individual has entered university after applied high school or after vocational training. For the higher vocational program I have left out latent types and for the lower vocational program I have left out both latent types and grades.

Duration risk Formula 3 shows the specification of duration risk for applied university and higher vocational programs. For the lower vocational program I left out grades. Applied high school and higher vocational training after lower vocational training have fixed lengths.

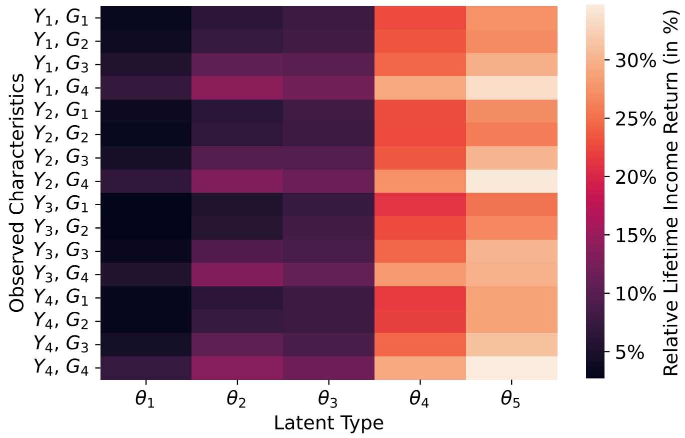
A.4. Additional figures

Figure A.3: Variation in applied high school enrollment rates across school types.



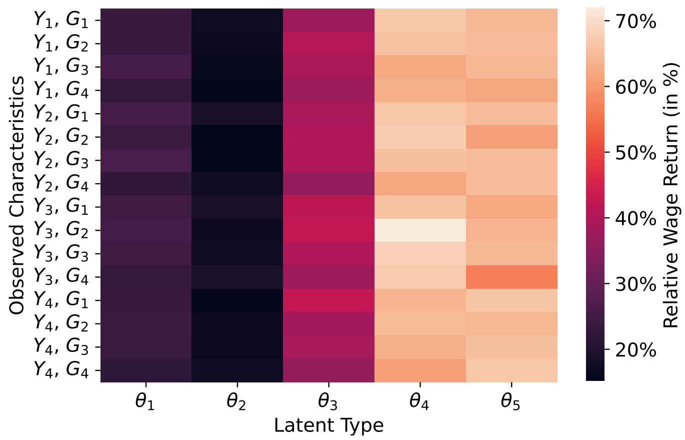
Note: This figure contains the enrollment rates at applied high school across different school types. See Subsection 3.3 for details on how school types are defined.

Figure A.4: Distribution of life-time earnings returns to applied university versus any alternative.



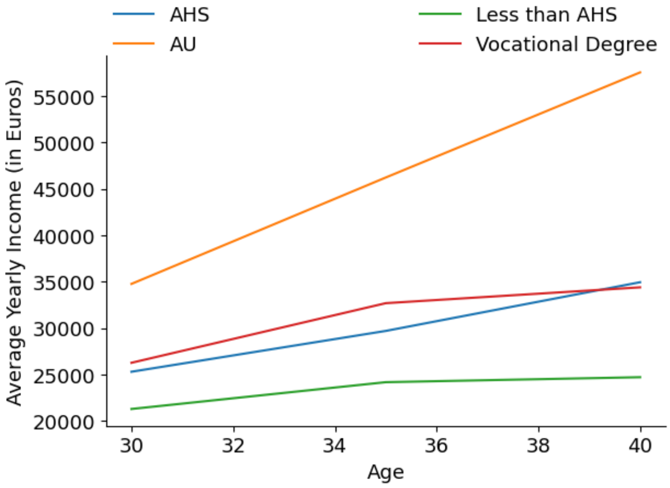
Note: This heatmap summarizes the distributions of returns to applied university versus any alternative. The vertical axis represents one combination of vocational school GPA and parental income, while the horizontal axis represents one latent type. The returns are expressed as percentages. The returns are obtained by calculating the difference in discounted lifetime income of individuals with and without applied university bachelor's degrees for each combination of observed characteristics and latent type. Observed characteristics are parental income and vocational school GPA. School type is not included since it has no direct effect on wages. Discounted lifetime income differentials within a group of observed characteristics and latent type are average returns to applied university for all individuals in that group since wages don't systematically differ conditional on these variables. The value of the respective group is then assigned to each simulated individual to obtain a distribution.

Figure A.5: Distribution of earnings returns to applied university at age 40.

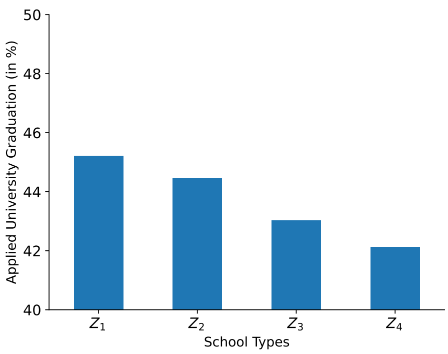


Note: This heatmap summarizes the distributions of returns to applied university at age 40. The vertical axis represents one combination of vocational school GPA and parental income, while the horizontal axis represents one latent type. The returns are expressed in percentage. The returns are obtained by calculating the difference in discounted lifetime income of individuals with applied bachelor's degrees and of individuals with higher vocational degrees for each combination of observed characteristics and latent type. Observed characteristics are parental income and vocational school GPA. School type is not included since it has no direct effect on wages. Discounted lifetime income differentials within a group of observed characteristics and latent type are average returns to applied university for all individuals in that group since wages don't systematically differ conditional on these variables. The value of the respective group is then assigned to each simulated individual to obtain a distribution.

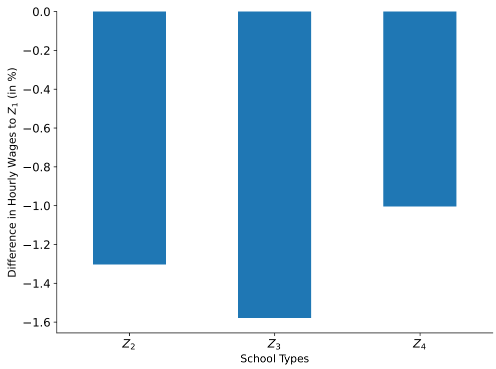
Figure A.6: Wage trajectories by educational attainment



Note: This figure shows the evolution of average hourly wages for individuals with and without applied university degrees. I only include individuals who work full-time. The applied university category only includes individuals with bachelor's degrees. The data is obtained from a cross section of hourly wages in 2019.



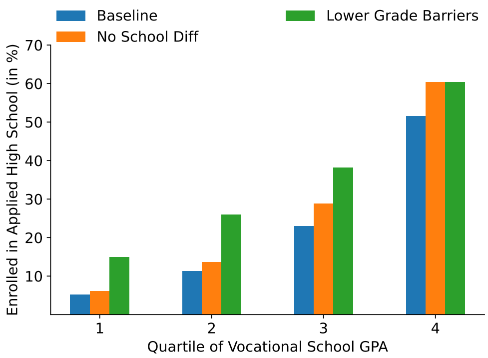
(a) Degree completion



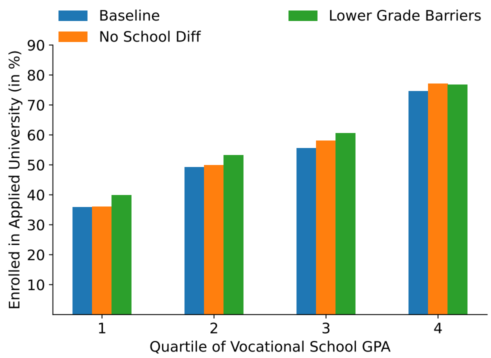
(b) Wages at age 28

Figure A.8: Variation across school types

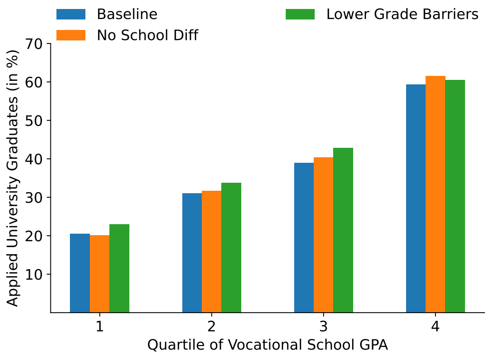
Note: This figure shows differences in conditional choices and outcomes across school types. Schools are grouped into four types based on their covariate-adjusted transition rate to applied high school as outlined in Subsection 3.3. The first panel shows differences in the probability of graduating from applied university, conditional on not having chosen the direct path to university for the highest quartile of vocational school GPA. The y-axis represents the proportion of graduates. The different school types are represented by different bars on the x-axis. The second panel shows differences in wages at age 28, conditional on having graduated from applied university via the alternative path. Wages are expressed relative to the school with the highest transition rate to applied high school. The different school types are represented by different bars on the x-axis.



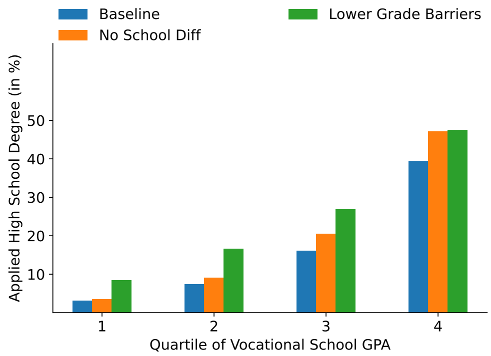
(a) Applied high school enrollment



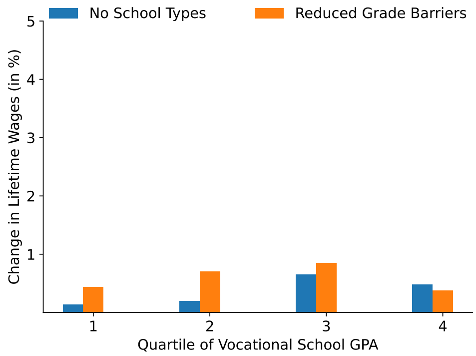
(b) Applied university enrollment



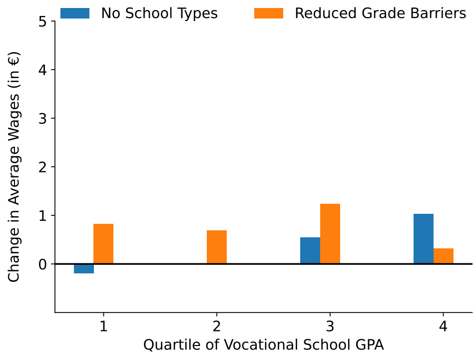
(c) Applied university degree



(d) Applied high school degree



(e) Lifetime wages



(f) Wages at age 40

Figure A.7: The effect of enforcing higher acceptance rates at high school

Note: This figure shows how enforcing higher acceptance rates at applied high school would affect enrollment and graduation at applied university and applied high school. Blue bars show the baseline model's proportions of applied university and applied high school graduates and enrollees. The orange bars show the proportions in the counterfactual scenario where all schools behave like the most lenient schools. The green bars show the proportions in the counterfactual scenario where people with lower vocational school GPA face lower barriers to enter applied high school and where all schools behave like the most lenient schools. The proportions are shown for each quartile of vocational school GPA. The last two panels show the effects on lifetime wages and wages at age 40 respectively.

References

Adda, J. and Dustmann, C. (2023). Sources of wage growth. *Journal of Political Economy*, 131(2):456–503.

Altonji, J. G. and Segal, L. M. (1996). Small-sample bias in gmm estimation of covariance structures. *Journal of Business & Economic Statistics*, 14(3):353–366.

Arcidiacono, P., Aucejo, E., Maurel, A., and Ransom, T. (2016). College attrition and the dynamics of information revelation. Technical report, National Bureau of Economic Research.

Bertrand, M., Mogstad, M., and Mountjoy, J. (2021). Improving educational pathways to social mobility: evidence from norway's reform 94. *Journal of Labor Economics*, 39(4):965–1010.

Birkelund, J. F. and van de Werfhorst, H. G. (2022). Long-term labor market returns to upper secondary school track choice: Leveraging idiosyncratic variation in peers' choices. *Social Science Research*, 102:102629.

De Groote, O. (2025). Dynamic effort choice in high school: Costs and benefits of an academic track. *Journal of Labor Economics*, 43(2):467–502.

Eckardt, D. (2023). Training specificity and occupational mobility: Evidence from german apprenticeships. Technical report, Technical report, mimeo, University of Warwick.

Fu, C. and Mehta, N. (2018). Ability tracking, school and parental effort, and student achievement: A structural model and estimation. *Journal of Labor Economics*, 36(4):923–979.

Gabler, J. (2022). A python tool for the estimation of large scale scientific models.

Hanushek, E. A., Schwerdt, G., Woessmann, L., and Zhang, L. (2017). General education, vocational education, and labor-market outcomes over the lifecycle. *Journal of human resources*, 52(1):48–87.

Heckman, J. J., Humphries, J. E., LaFontaine, P. A., and Rodriguez, P. L. (2012). Taking the easy way out: How the ged testing program induces students to drop out. *Journal of labor economics*, 30(3):495–520.

Joensen, J. S. and Mattana, E. (2021). Student aid design, academic achievement, and labor market behavior: grants or loans? *Academic Achievement, and Labor Market Behavior: Grants or Loans.*

Keane, M. P. and Wolpin, K. I. (1997). The career decisions of young men. *Journal of political Economy*, 105(3):473–522.

Matthewes, S. H. and Ventura, G. (2022). On track to success?: Returns to vocational education against different alternatives.

McNally, S., Ventura, G., and Virtanen, H. (2022). Returns to vocational education and training. In *Handbook of labor, human resources and population economics*, pages 1–30. Springer.

Mountjoy, J. (2022). Community colleges and upward mobility. *American Economic Review*, 112(8):2580–2630.

Powell, M. J. et al. (2009). The bobyqa algorithm for bound constrained optimization without derivatives. *Cambridge NA Report NA2009/06, University of Cambridge, Cambridge*, 26.

Proctor, A. (2022). Did the apple fall far from the tree?

Silliman, M. and Virtanen, H. (2022). Labor market returns to vocational secondary education. *American Economic Journal: Applied Economics*, 14(1):197–224.

Stinebrickner, T. and Stinebrickner, R. (2012). Learning about academic ability and the college dropout decision. *Journal of Labor Economics*, 30(4):707–748.

Tincani, M. M., Kosse, F., and Miglino, E. (2023). *College access when preparedness matters: New evidence from large advantages in college admissions*. Centre for Economic Policy Research.

Todd, P. E. and Wolpin, K. I. (2023). The best of both worlds: combining randomized controlled trials with structural modeling. *Journal of Economic Literature*, 61(1):41–85.

Van Esch, W. and J., N. (2010). Van vmbo naar havo: tweestrijd over tweesprong? Technical report.

van Esch, W. and Neuvel, J. (2009). Een stukje van de nederlandse droom: doorstroom van vmbo naar havo. Technical report, Expertisecentrum Beroepsonderwijs (ecbo), 's-Hertogenbosch. Bestelnummer A00539.

Zhu, Z. (2025). Effects of free community colleges: Evidence from a dynamic school choice model. Unpublished manuscript.

Zumbuehl, M., Chehber, N., Dillingh, R., et al. (2022). Can skills differences explain the gap in track recommendation by socio-economic status? Technical report, CPB Netherlands Bureau for Economic Policy Analysis.